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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

MTH 112 — Past Questions & Materials
by Renance

Course: MTH 112
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 2
Levels: 100

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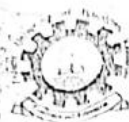
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Paper 1 of 2 — MTH 112 Past Question

Level 100 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM & MINERALS
COLLEGE OF SCIENCE
DEPARTMENT OF MATHEMATICS / COMPUTER SCIENCE
FIRST SEMESTER EXAMINATION 2017/2018

Level: 100

Course Code: MTH 112

Title: Data Analysis

Duration: 2 hrs

Name: Matric No: S/N on Attendance:
Instruction: Answer all questions by shading the correct option in the question paper and copy the same option on the first page of the booklet. Submit both the question paper and the booklet for marking.

TYPE B

Use the following distribution below for questions 1 to 3.
The following data shows the number of hours worked by 250 Maths/Computer Sc Students

Number of Hours	0 - 9	10 - 19	20 - 29	30 - 39	40 - 49	50 - 59
Frequency	40	50	70	40	30	20

- What is the mode of the above distribution? (A) 17.5 (B) 29.5 (C) 26.5 (D) 23.5
- What is the first quartile of the above distribution? (A) 12 (B) 14 (C) 16 (D) 18
- What is the median of the above distribution? (A) 20.5 (B) 23.5 (C) 24.5 (D) 25.5
- The arithmetic mean of a set of 5 numbers is 20. If each number is first multiplied by 2 and then added by 5, then what is the mean of new numbers? (A) 11.5 (B) 22.5 (C) 33.5 (D) 44.5
- What is the weighted mean of first 10 natural numbers whose weights are equal to the corresponding number? (A) 7 (B) 5.5 (C) 5 (D) 4.5
- The sum of mean, mode and median of the data 12, 15, 11, 13, 18, 11, 13, 12, 13 to the nearest whole number is (A) 39 (B) 36 (C) 37 (D) 38
- The arithmetic mean of the first ten whole numbers is (A) 5.5 (B) 5 (C) 4 (D) 4.5
- If the arithmetic mean of the following data $6 - x, 7 + 2x, 3 - 3x, 5 + 4x, 4 - 5x, 11 + 6x, 2 - 7x, 18 + 0x$ is 10, what is the value of x ? (A) 5 (B) 6 (C) 7 (D) 4
- What is the geometric mean of the following data 4, 7, 3, 5, 3, 6, 4, 4, 5, 4, 6, 6, 8? (A) 4.33 (B) 4.79 (C) 4.46 (D) 4.70

Use the following distribution below for questions 10-15

Class Interval	13 - 17	18 - 22	23 - 27	28 - 32	33 - 37	38 - 42	43 - 47	48 - 52	53 - 57
Frequency	2	22	10	14	3	4	6	1	1

- Find the standard deviation of the above distribution (A) 19.40 (B) 16.30 (C) 9.40 (D) 11.39
- What is the variance? (A) 88.45 (B) 93.60 (C) 77.40 (D) 69.70
- What is the coefficient of variation? (A) 33.27 (B) 26.14 (C) 39.70 (D) 63.40
- Determine the degree of asymmetry of the distribution (A) 0.0617 (B) 0.0120 (C) -0.0617 (D) -0.0566
- What is the cumulative frequency of 43 - 47 class interval? (A) 6 (B) 57 (C) 61 (D) 63
- What is the cumulative frequency before the third quartile class? (A) 34 (B) 48 (C) 24 (D) 51
- What is the harmonic mean of the following data 4, 7, 3, 5, 3, 6, 4, 4, 5, 4, 6, 6, 8? (A) 4.23 (B) 4.58 (C) 4.74 (D) 4.30
- Suppose that the mean and median of a given distribution are 63.7 and 62.5 respectively, determine the mode. (A) 60.1 (B) 63.7 (C) 62.5 (D) 61.2
- Find the mean deviation of the following data 3, 5, 6, 7, 8, 6, 10, 12 (A) 2.12 (B) 2.16 (C) 2.30 (D) 2.40
- The sum of the percent frequencies for all classes will always equal to (A) one (B) the number of classes (C) the number of items in the study (D) 100
- Since the mode is the most frequency occurring data value, then (A) it can never be larger than the mean (B) it is always greater than the mean (C) it is always greater than the median (D) none of the above
- If the semi-interquartile range value and third quartile values are 12.9 and 66 respectively, find the value of the first quartile (A) 30.2 (B) 40.2 (C) 20.3 (D) 35.6

$$\frac{S.D}{\text{mode}} = \frac{9.4}{60.1}$$

$$\frac{\text{mean} - \text{mode}}{S.D} = \frac{63.7 - 60.1}{10}$$

$$28.55$$

$$9.4$$

22. A series of numerical figures which show the relative position is called (A) Index number (B) Relative number (C) Absolute number (D) None of the above
23. Index number for the base period is always taken as (A) 200 (B) 50 (C) 1 (D) 100
24. Index numbers are often constructed from the (A) frequency (B) Class (C) sample (D) None
25. The ratio of price of single commodity in a given period to its price in another period is called the (A) Base period (B) Price ratio (C) Relative price (D) None
26. _____ is a good example of sampling frame. (A) Voter's card (B) Voter's register (C) Class attendance (D) Employee's name
27. Why is sample preferred to population in most statistical enquiry? (A) Biased (B) Statistic (C) Unbiased (D) Parameter
28. Select a sample of 5 members from a population consisting of 20 members. Given that the first number selected is 3. Find the mean of the sample using systematic random sampling (A) 11 (B) 3 (C) 7 (D) 19
29. Which of the following is a quasi-homogeneous random sampling? (A) Systematic random sampling (B) Simple random sampling (C) Stratified random sampling (D) Cluster sampling
30. _____ is a sampling method drawn from different groups from a given population. (A) Multi-stage sampling (B) Stratified random sampling (C) Cluster sampling (D) Systematic random sampling
31. A survey was carried out on some students of the MTH 112 to assess their performance. What is this type of sampling called? (A) Simple random sampling (B) Cluster sampling (C) Multi-stage sampling (D) Pilot survey
32. A measurement scale that serve as a label is term (A) Ordinal (B) Ratio (C) Interval (D) Nominal
- Given the follow sample distribution table
- | | | | | |
|----------------|-------|-------|-------|-------|
| Class interval | 31-40 | 41-50 | 51-60 | 61-70 |
| Frequency | 6 | 10 | 3 | 1 |
33. What is the upper class boundary of the second class (A) 40.5 (B) 50.5 (C) 60.5 (D) 30.5
34. The following are marks scored by some students in 100 level
14, 19, 18, 12, 18, 8, 19, 16, 10, 11, 3, 17, 18, 11, 15, 5, 20, 16, 12, 3.
Find the frequency mark for the class interval 16-20, using the class size of 5. (A) 4 (B) 2 (C) 9 (D) 8
35. Which of the following is a common factor to bar chart and histogram (A) Class interval (B) Class boundary (C) Frequency (D) Class marks
36. A chemist has undertaken an experiment where he tried adding 6 different amounts of a catalyst in a mixture given as 4.1, 2.0, 6.0, 2.8, 1.1, and 4.9. Find $\sum_{i=1}^6 (x_i - \bar{x})$: (A) 20.9 (B) 3.0 (C) 6.97 (D) 0
37. Which of the following is the variance of y ? (A) $\sum_{i=1}^n y_i^2 - \bar{y}$ (B) $\sum_{i=1}^n y_i^2 - \sum_{i=1}^n y_i$ (C) $\sum_{i=1}^n (y_i - \bar{y}) y_i$ (D) $\sum_{i=1}^n y_i^2 - \bar{y}^2$
38. Given the following data 60, 49, 40, 61, 64, 60, 59, 54, 62, 69, 70, 42, 56, 61, 61, 61, 58, 51, 48, 65, 49, 49, 41, 48, 52, 46, 59, 46, 58, 43. What is the sector angle of 69? (A) 4 (B) 10.8 (C) 14.4 (D) 7.2
39. Find the semi-interquartile range of 10 students scores in PHY 109 given below 10, 6, 6, 2, 10, 3, 7, 4, 8, 4 (A) 3.75 (B) 8.5 (C) 4.75 (D) 2.38
40. Which of the following best explain why we classify data? (A) to prepare tabulation (B) to eliminate necessary details (C) to draw inference (D) to present data in minimum space
41. If $\alpha = 1.5\%$, what is the table value for a two-tailed large sample test.
(A) 1.96 (B) 2.43 (C) 2.17 (D) 1.645
42. A test of the degree of association based on ranking is carried out using (A) regression (B) Spearman's method (C) Pearson's method (D) product moment.
43. Given the fitted regression model $y = -20 - 4x$. The relationship between x and y is (A) positive (B) direct (C) inverse (D) equal
44. If the correlation coefficient is -0.2 , then the relationship between the variables is (A) low direct (B) low inverse (C) high direct (D) high inverse
45. The sample size required to estimate the mean if the error is to be less than 1000 with 95% level of confidence and standard deviation 10000 is (A) 384 (B) 19 (C) 20 (D) 385
46. A sample of 75 renters in a city showed that 40 were under the age of 35. Construct a 99% confidence interval for the percentage of all renters that are under the age of 35. (A) (0.38, 0.68) (B) (0.48, 0.88) (C) (0.36, 0.97) (D) (0.42, 0.92)

1550 = 30

(2+3) =

47. Given that the calculated value for sample mean is 10 and the critical test at $\alpha = 1\%$ is $Z = 2.33$. The P-value for the test is: (A) 0.27 (B) 0.73 (C) 0.54 (D) 0.74
48. Given that the calculated value for large sample one-tailed test at $\alpha = 1\%$ is $Z = -2.21$. The P-value for the test is: (A) 0.05 (B) 0.01 (C) 0.97 (D) 0.99
49. The variance of a set of data is 12. Find the new standard deviation if each of the observation is multiplied by 2. (A) 4.80 (B) 2.40 (C) 2.19 (D) 1.55
50. The standard deviation of a set of data is 2.5. Find the new standard deviation if each of the observation is divided by 4. (A) 1.25 (B) 1.56 (C) 0.39 (D) 0.63
51. An estimate of population parameter that is based on sample is derived using sample: (A) parameter (B) mean (C) statistic (D) variance
52. The sample size required for large sample inference is: (A) $n \geq 30$ (B) $n \leq 30$ (C) $n \geq 100$ (D) $n \geq 30$
53. An estimator whose expected value equals the estimated value is said to be (A) unbiased (B) biased (C) correct (D) appropriate
54. The two conditions that lead to the use of small sample test statistic are (A) $n < 30$ and σ known (B) $n < 30$ and σ unknown (C) $n \geq 30$ and σ unknown (D) None of the above
55. Using the p-value, the null hypothesis is rejected if: (A) $p < \alpha$ (B) $p > \alpha$ (C) $p = \alpha$ (D) $p \geq \alpha$
56. The statistical distribution appropriate for small sample test of significant difference in means is (A) Normal (B) Chi-Square (C) Student's t (D) Fisher F
57. Rejection of the null hypothesis when it is true leads to: (A) type I error (B) one-tailed (C) type II (D) two-tailed
58. In test of hypothesis, the second step is (A) calculated value of the test statistic (B) H_0 and H_1 (C) critical region (D) decision
59. If $n = 144$, $\bar{x} = -2.1$, $\sigma = 1.2$. A 95% confidence interval for μ is: (A) (1.91, 2.30) (B) (-2.30, -1.91) (C) (-2.26, -1.94) (D) (1.94, 2.26)
60. If $n = 25$, $\bar{x} = -2.1$, $S = 1.1$. A 95% confidence interval for μ is: (A) (1.86, 2.34) (B) (-2.34, -1.86) (C) (1.60, 2.60) (D) (-2.60, -1.60)
- Use the following information to answer questions 61 to 62.
If $\bar{x}_1 = 6.5$, $S_1 = 2.5$, $n_1 = 100$, $\bar{x}_2 = 8.5$, $S_2 = 2.6$, $n_2 = 200$.
61. The value of the test statistic for testing the significant difference between the two sample means is: (A) -6.45 (B) 6.45 (C) -10.26 (D) 10.26
62. What is the decision for the hypothesis using $\alpha = 5\%$? (A) Accept H_0 (B) Cannot reject H_0 (C) Reject H_0 (D) Reserve judgment
- Use the following information to answer questions 63 to 66.
If $\bar{x}_1 = 5$, $S_1 = 1.2$, $n_1 = 25$, $\bar{x}_2 = 3$, $S_2 = 1.5$, $n_2 = 16$.
63. The pooled standard deviation for the two samples is (A) 1.32 (B) 1.75 (C) 4.73 (D) 3.57
64. The value of the test statistic for testing the significant difference between the two sample means is: (A) 3.57 (B) 1.75 (C) 4.73 (D) 1.32
65. What is the table value for testing the hypothesis using $\alpha = 1\%$. (A) 1.685 (B) 2.023 (C) 2.426 (D) 2.708
66. What is the decision for the hypothesis using $\alpha = 1\%$? (A) Accept H_0 (B) Cannot reject H_0 (C) Reject H_0 (D) Reserve judgment
67. A bakery claims that the weight of its bread is more than 500g. To test if the actual weight is less than the company claim, samples were selected and hypothesis testing carried out. What type of test is this? (A) type I (B) type II (C) two-tailed (D) one-tailed
- Use the following information to answer questions 68 to 70.
A 95% confidence interval reported for the parameter μ is 1.5 to 6.5. The interval is based on 15 observations.
68. What is the table value of the t-distribution? (A) 2.145 (B) 2.624 (C) 1.761 (D) 2.977
69. What is the value of the standard error of mean (s.e. $(\bar{x})$)? (A) 1.19 (B) 1.17 (C) 1.96 (D) 1.86
70. What is the mean of the unknown data set? (A) 8.0 (B) 12.0 (C) 2.5 (D) 4.0



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DEPARTMENT OF MATHEMATICS / COMPUTER SCIENCE
FIRST SEMESTER EXAMINATION 2016/2017
Course Code: MTH 112
Level: 100
Title: Data Analysis
Duration: 2 $\frac{1}{2}$ hrs

Name: Matric No: S/N on Attendance:
Instruction: Answer all questions by shading the correct option in the question paper and copy the same option on the first page of the booklet. Submit both the question paper and the booklet for marking.

TYPE A

- Rejection of the null hypothesis when it is true leads to error
(a) type I (b) type II (c) one-tailed (d) two-tailed
- The critical region is the region set aside for
(a) accepting H_0 (b) rejecting or accepting H_0 (c) rejecting H_0 (d) reserving judgment
- The critical value for the rejection of the null hypothesis for large sample one-tailed test at $\alpha = 1\%$ is (a) 1.96
(b) 1.38 (c) 1.64 (d) 2.33
- If $n = 225$, $\bar{x} = -5.2$, $\sigma = 2.5$. A 99% confidence interval for μ is
(a) (4.77, 5.63) (b) (-5.29, -4.91) (c) (-5.63, -4.77) (d) (-6.91, -7.29)
Use the following information to answer questions 5 to 7.
If $\bar{x}_1 = 10.5$, $S_1 = 12.5$, $n_1 = 144$, $\bar{x}_2 = 8.5$, $S_2 = 11.5$, $n_2 = 100$.
- The calculated value of the test statistic for testing the significant difference between the two sample means is
(a) 1.96 (b) 1.29 (c) 5.62 (d) 2.58
- What is the decision for the hypothesis using $\alpha = 5\%$
(a) Accept H_0 (b) Reject H_0 (c) Reserve judgment (d) None of the above
- The P-value for the hypothesis using $\alpha = 5\%$ is
(a) 1.96 (b) 0.582 (c) 0.197 (d) 1.64
Use the following information to answer questions 8 to 10.
If $\bar{x}_1 = 5$, $S_1 = 0.5$, $n_1 = 10$, $\bar{x}_2 = 5.5$, $S_2 = 0.25$, $n_2 = 15$.
- The calculated value of the test statistic for testing the significant difference between the two sample means is
(a) 1.96 (b) 2.36 (c) 5.36 (d) 3.32
- What is the table value for testing the hypothesis using $\alpha = 1\%$
(a) 2.807 (b) 2.797 (c) 2.500 (d) 2.492
- The hypothesis decision is (a) Reserve judgment (b) Accept H_0 (c) Reject H_0 (d) None of the above
- If $n = 25$, $\bar{x} = 15.15$, $S = 1.5$. A 95% confidence interval for μ is
(a) (14.56, 15.74) (b) (14.53, 15.77) (c) (6.51, 7.69) (d) (16.69, 17.51)
- A company claims that its bag of rice weight 50kg. To test if the actual weight is less than the company claim, samples were selected and hypothesis testing carried out. What type of test is this
(a) type I (b) type II (c) two-tailed (d) one-tailed
- H_0 for testing hypothesis using normal distribution is rejected if
(a) $Z_{cal} < Z_{tab}$ (b) $P - value < \alpha$ (c) $Z_{cal} \leq Z_{tab}$ (d) $P - value > \alpha$
- If $\alpha = 7.5\%$, what is the table value for a two-tailed test using normal distribution
(a) 1.98 (b) 1.64 (c) 1.78 (d) 1.76
- Determine the P-value for a one-tailed test if the calculated value of the normal distribution statistic is -1.72 (a) 0.0427 (b) 1.0962 (c) 0.9573 (d) 1.0952
- The time taken by a typist to complete a task on a machine is normally distributed with $\sigma = 2.5$ mins. In a sample of 25 tasks, $\bar{x} = 25.5$ mins was found. Construct a 95% confidence interval for μ
(a) (25.64, 26.96) (b) (24.52, 26.48) (c) (21.64, 26.96) (d) (24.52, 26.75)
- The 99% confidence interval for μ given a data set with $n = 36$, $\bar{x} = 15.5$, $S = 0.75$ is
(a) (13.50, 15.58) (b) (15.50, 15.58) (c) (15.18, 15.82) (d) (15.40, 15.50)
Use the following information to answer questions 18 to 20. A 95% confidence interval reported for the parameter μ is 5.27 to 14.73. The interval is based on 5 observations constructed using t-distribution.
- What is the table value of the t-distribution (a) 2.131 (b) 2.776 (c) 2.582 (d) 1.642
- What is the value of the standard error of mean ($s.e(\bar{x})$) (a) 6.80 (b) 6.48 (c) 1.70 (d) 4.50
- What is the mean of the unknown data set (a) 10.00 (b) 11.50 (c) 12.50 (d) 20.50
Use the following information to answer questions 21 to 23.
The mean score in an examination from a random sample of 25 students is 47.50 with $S = 2.50$.
- The calculated value of the statistic for testing the claim that the mean score is higher than 45 at $\alpha = 1\%$
(a) 2.81 (b) 2.58 (c) 5.64 (d) 5.00
- The table value for the test is (a) 1.96 (b) 2.49 (c) 1.64 (d) 2.58

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23. The hypothesis decision is (a) Reserve judgment (b) Accept H_0 (c) Reject H_0 (d) None of the above
24. The variance of a set of data represented by Y is 1.50 then the variance of $(Y - 2)$ is (a) 6.0 (b) 1.5 (c) 3.0 (d) 4.5
25. X represents a set of data with variance 5.0, the variance of $(\frac{X}{5})$ is (a) 1.25 (b) 2.50 (c) 5.0 (d) 7.5
- Use the following information to answer questions 26 to 30. A random sample of 36 Shoprite customers spent a mean amount of 450.50 naira and standard deviation of 100.50 naira.
26. The standard error of means for this sample is (a) 6.75 (b) 1.96 (c) 1.65 (d) 16.75
27. The calculated value of the statistic for testing the claim that the mean amount spent is less than 400 at 0.05 level of significance is (a) 1.03 (b) 3.01 (c) 3.50 (d) 1.11
28. The table value for this test is (a) 1.96 (b) 2.33 (c) 1.65 (d) 2.58
29. The P-value for this test is (a) 0.0013 (b) 2.0313 (c) 0.6523 (d) 2.5823
30. The hypothesis decision is (a) Reserve judgment (b) Reject H_0 (c) Accept H_0 (d) None of the above
- Use the following information to answer questions 31 to 35. The following are the test scores of students 30, 20, 16, 10, 24.
31. The unbiased estimate of the population standard deviation is (a) 6.81 (b) 11.60 (c) 3.81 (d) 7.62
32. The standard error of mean for the sample is (a) 6.80 (b) 3.05 (c) 3.41 (d) 4.50
33. The calculated value of the statistic for testing the claim that the sample mean is not significantly different from 22 at $\alpha = 5\%$ is (a) -0.59 (b) 3.01 (c) -3.50 (d) 0.59
34. The critical value used for the test in question 33 is (a) 1.96 (b) 2.58 (c) 2.78 (d) 1.64
35. The hypothesis decision is (a) Reject H_0 (b) Accept H_0 (c) Reserve judgement (d) None of the above
36. Let $\{(1,5), (3,4), (m,2), (4,5), (5,1), (1,3), (1,4), (2,5), (3,5), (0,6)\}$ be points on the Euclidean plane with equal means. Find the sample variance of x . (a) 20 (b) 19 (c) 34 (d) 6
37. Find the value of $\text{var}(X - Y)$ if the $\text{var}(X)$, $\text{var}(Y)$ and $\text{cov}(X, Y)$ are $2\frac{1}{2}$, 4, and $\frac{1}{2}$ respectively. (a) 7 (b) $6\frac{1}{2}$ (c) 3 (d) 6
38. The test records for mathematics and computer students are presented in pair respectively (1,2), (3,3), (3,5), (5,4), (5,6), (6,5), (8,7), (9,8). Find the $\text{cov}(X, Y)$. (a) 3.50 (b) 6.25 (c) 4.24 (d) 5.00
39. Find the coefficient of variation for computer students of test records for Mathematics and Computer students (1,2), (3,3), (3,5), (5,4), (5,6), (6,5), (8,7), (9,8). (a) 50 (b) 70 (c) 37 (d) 125
40. Find the $\text{cov}(20, a)$ if a is a constant of variation whose probability of occurrence is one. (a) 20 (b) 400a (c) 0a (d) a
41. If the $\text{var}(X) = 15$. Find the $\text{cov}(aX, X)$ if X is a random variable of a population and $a = 2$ (a) 15 (b) 225 (c) 30 (d) 15a
42. What is the value of x if the $2x - 1$ is the sectorial angle of $1/12$ with a pie-chart whole cumulative is 12 (a) 2 (b) 1 (c) $1\frac{1}{2}$ (d) 12
43. Find the $\text{var}(X - Y)$ if X and Y are independently and identically distributed if the $\text{var}(X)$, $\text{var}(Y)$ and $\text{cov}(X, Y)$ are $2\frac{1}{2}$, 4, and $\frac{1}{2}$ respectively. (a) 6 (b) 7 (c) $6\frac{1}{2}$ (d) 3
44. When frequency data and count data are represented by counts or number of observations in each group of observation in a population, they are referred to as --- data (a) categorical (b) measurement (c) quantitative (d) parameter
45. --- assigns numbers to object of a sample in an experiment. (a) measurement (b) parameter (c) statistic (d) nominal scale
46. Find the mean of the distribution 66, 75, 72, 71, 55, 56, 72, 93, 72, 72, 73, 73, 91, 66, 71, 56, 59. (a) 66.4 (b) 72.00 (c) 66.3 (d) 70
47. Which of the following represent the characteristics of a good questionnaire? (a) Tele-guiding questions (b) Long and explanatory questions (c) Honest and vague questions (d) Unambiguous
- The table below represents the examination scores of 100 levels students in MTH 112 in 2013/2014 session with mean 18.38. Use the table to answer questions 48 to 50
- | | | | | | |
|-----|--------|---------|---------|---------|---------|
| x | 6 - 10 | 11 - 15 | 16 - 20 | 21 - 25 | 26 - 30 |
| f | 7 | 7 | 11 | 10 | 6 |
48. Find the variance of the distribution (a) 6.00 (b) 18.38 (c) 41.52 (d) 6.40
49. Find the value of y in the distribution (a) 6.0 (b) 18.00 (c) 41.00 (d) 8.00
50. What is the median of the distribution? (a) 6.0 (b) 18.3 (c) 41.00 (d) 21.00
51. Find the mode of 66, 75, 72, 71, 55, 91, 66, 71, 56, 59, 56, 72, 93, 72, 72, 73, 73 (a) 66 (b) 72 (c) 56 (d) 73
- Use the information to answer questions 52 - 55
- | | | | | | | | | |
|-----|-----|-----|-----|-----|-----|-----|-----|---|
| x | 1.2 | 1.3 | 1.4 | 2.4 | 0.8 | 3.4 | 0.2 | 1 |
| y | 5 | 6 | 1 | 2.9 | 4.2 | k | 5 | 3 |
52. What is the value of k if the slope is 1.327? (a) 7 (b) 3 (c) 10 (d) 8
53. Find the intercept if $y = a + bx$ is the regression equation (a) 3 (b) 7 (c) 8 (d) 10

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54. Which of the equation represents the regression equation? (a) $y = 3 + 1.3x$ (b) $y = 7 + 1.8x$ (c) $y = 2 - 1.8x$ (d) $y = 4.2 + 8x$

55. What is the mean of the variable y ? (a) 4.64 (b) 1.46 (c) 10.00 (d) 1.00

56. What is the value of y if the mean of the number $13, 2y + 1, 0, 5y - 1$, and 11 is 9 ? (a) 1.7 (b) 3.0 (c) 4.0 (d) 7.0

57. The mean and the median of a symmetric distribution is said to be equal when they have ----- distribution? (a) unimodal (b) bimodal (c) multimodal (d) mode

58. The staff strength of Jom-oil Company is shown below. Find the median.

Staff strength	1-10	11-20	21-30	31-40	41-50	51-60	61-70	71-80	81-90	91-100
No. of company	1	5	10	19	42	10	6	4	2	1

(a) 44.07 (b) 42.00 (c) 77.00 (d) 45.00

59. The time taken in seconds by 100 different chemical substances to melt when subjected to a particular temporary condition are given below. Find the mode

Time (sec)	4.51 - 5.32	5.33 - 6.14	6.15 - 6.96	6.97 - 7.78	7.79 - 8.60	8.61 - 9.42
frequency	15	7	35	28	10	5

(a) 6.145 (b) 6.801 (c) 6.965 (d) 35

60. Find the probability of the class interval 30 - 39 in the table below

x	0-9	10-19	20-29	30-39	40-49	50-59
y	10	2	4	5	25	14

(a) 1 (b) $3/10$ (c) $1/5$ (d) $1/12$

61. What is the standard deviation of the distribution for an average of 48.28?

X	25	29	$y + 3$	59	70
f	7	9	8	6	20

(a) 366.96 (b) 18.96 (c) 359.63 (d) 19.16

62. Find the value of y for an average of 48.28 in the table below

X	$y + 3$	70	25	59	29
f	8	20	7	6	9

(a) 38 (b) 50 (c) 40 (d) 25

63. Given the table below for 100 level 50 test scripts

11-15	16-20	21-25	26-30	31-35	36-40
6	5	x	12	9	5

If the class intervals are represented in a pie chart, what is the sectorial angle of the class interval 26-30? (a) 43.2 (b) 86.4 (c) 93.6 (d) 64.8

64. Find the class size of this distribution

x	26-30	31-35	36-40	41-45
frequency	p	y	10	21

(a) 15.5 (b) 5.0 (c) 25.6 (d) 30.5

65. The normal distribution is represented by

(a) $\frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\sigma^2(x-\mu)^2}$ (b) $\frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}\sigma(x-\mu)^2}$ (c) $\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\sigma(x-\mu)^2}$ (d) $\frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$

Use the information below to answer question 66 and 67

x	15-19	20-24	25-29	30-34	35-40
f	3	4	7	5	1

66. Find the mean of the distribution (a) 27.50 (b) 26.25 (c) 7.00 (d) 30.00

67. Find the skewness (a) 0.40 (b) 2.25 (c) 0.22 (d) -0.40

68. The table below shows a frequency distribution for grouped data. Calculate the mode

x	0-5	5-15	15-25	25-35	35-45	45-100
frequency	2	6	20	15	8	4

(a) 24.75 (b) 22.37 (c) 27.73 (d) 27.50

69. Find the sum of the deviation from the mean of the distribution 4, 6, 2, 8, 10, 3, 6, 4, 7, 10. (a) 22 (b) 6 (c) 0 (d) 7.8

70. ----- is a numerical value that summarizes population

(a) Parameter (b) Statistic (c) Inferential statistic (d) Descriptive statistic

Best of Luck.

Examiners: Dr. A. M. Okedoye, S. O. Ogundele, J. T. Eghwerido.



FEDERAL UNIVERSITY OF PETROLEUM RESOURCES EFFURUN
COLLEGE OF SCIENCE
DEPARTMENT OF MATHEMATICS / COMPUTER SCIENCE
FIRST SEMESTER EXAMINATION 2015/2016

Course Code: MTH 112
Level: 100

Title: Data Analysis
Duration: 2 hrs

TYPE A

Section A: Answer all questions in this section.

The question given to me is for Type ~~A~~ -----

(A) (B) (C) (D)

Section B: Answer all questions in this section.

1. Academic class of degree (First, Second, Third) is an example of ----- scale
(a) nominal (b) ~~ordinal~~ (c) interval (d) ratio
 2. Acceptance of the null hypothesis when it is false leads to ----- error
(a) type I (b) ~~type II~~ (c) one-tailed (d) two-tailed
 3. The first step in hypothesis testing is the determination of -----
(a) critical region (b) ~~H_0 and H_1~~ (c) calculated value of the test statistic (d) decision
 4. If $n = 148$, $\bar{x} = -7.1$, $\sigma = 1.2$. A 95% confidence interval for μ is -----
(a) (6.91, 7.29) (b) (-7.29, -6.91) (c) (-7.29, 6.91) (d) ~~(-6.91, -7.29)~~
 5. If $n = 16$, $\bar{x} = -7.1$, $S = 1.1$. A 95% confidence interval for μ is -----
(a) (6.51, 7.69) (b) ~~(-7.69, -6.51)~~ (c) (-6.51, 7.69) (d) (6.69, 7.51)
- Use the following information to answer questions 6 to 7.
If $\bar{x}_1 = 8.5$, $S_1 = 2.3$, $n_1 = 100$, $\bar{x}_2 = 7.5$, $S_2 = 2.8$, $n_2 = 110$.
6. The value of the test statistic for testing the significant difference between the two sample means is -----
(a) 1.96 (b) -2.84 (c) 2.84 (d) 2.58
 7. What is the decision for the hypothesis using $\alpha = 5\%$ -----
(a) Accept H_0 (b) Cannot reject H_0 (c) Reject H_0 (d) Reserve judgment
- Use the following information to answer questions 8 to 10.
If $\bar{x}_1 = 3$, $S_1 = 0.9$, $n_1 = 50$, $\bar{x}_2 = 3.5$, $S_2 = 1.2$, $n_2 = 50$.
8. The value of the test statistic for testing the significant difference between the two sample means is -----
(a) ~~1.96~~ (b) 2.36 (c) -2.36 (d) -1.107
 9. What is the table value for testing the hypothesis using $\alpha = 1\%$ -----
(a) 1.96 (b) 1.64 (c) ~~2.58~~ (d) 2.24
 10. What is the decision for the hypothesis using $\alpha = 1\%$ -----
(a) ~~Reserve judgment~~ (b) Cannot accept H_0 (c) Reject H_0 (d) Accept H_0
 11. A company claims that its meat pie takes 10 minutes to make. To test if the actual time taken is more than the company claim, samples were selected and hypothesis testing carried out. What type of test is this
(a) type I (b) type II (c) two-tailed (d) ~~one-tailed~~
 12. H_0 for testing hypothesis using normal distribution is rejected if
(a) $Z_{cal} < Z_{tab}$ (b) $Z_{cal} \leq Z_{tab}$ (c) $Z_{cal} \geq Z_{tab}$ (d) ~~$Z_{cal} > Z_{tab}$~~
 13. If $\alpha = 2.5\%$, what is the table value for a two-tailed test using normal distribution
(a) 1.96 (b) 1.64 (c) 2.58 (d) ~~2.24~~
 14. The critical region is the region for
(a) Acceptance of H_0 (b) Acceptance of H_1 (c) Rejection of H_1 (d) ~~Rejection of H_0~~
 15. The use of sample for taking decision about the population is refer to as ----- statistic.
(a) Descriptive (b) Estimation (c) Sampling (d) ~~Inferential~~
 16. The volume of wine filled by a bottling machine is normally distributed with $\sigma = 5$ cl. In a sample of 16 bottles, $\bar{x} = 74.3$ cl was found. Construct a 95% confidence interval for μ
(a) ~~(71.64, 76.96)~~ (b) (71.85, 76.75) (c) (-71.64, 76.96) (d) (-71.85, 76.75)
 17. The 95% confidence interval for a data set with $n = 34$, $\bar{x} = 3.54$, $S = 0.13$ is
(a) ~~(3.50, 3.58)~~ (b) (-3.50, 3.58) (c) (3.40, 3.50) (d) (-3.40, 3.50)
- Use the following information to answer questions 18 to 20.

A 95% confidence interval reported for the parameter μ is 1.6 to 2.8. The interval is based on 16 observations constructed using t-distribution.

18. What is the table value of the t-distribution
(a) 2.131 (b) 1.64 (c) 2.58 (d) 2.24

19. What is the value of the standard error of mean ($s.e(\bar{x})$)
(a) 1.45 (b) 2.131 (c) 4.18 (d) 1.64

20. What is the mean of the unknown data set
(a) 4.70 (b) 1.60 (c) 1.50 (d) 2.54

Use the following information to answer questions 21 to 23.

The mean price of a dozen roses at Shoprite from a random sample of size 30 is 48.33 with $S = 6.07$.

21. The calculated value of the statistic for testing the claim that the mean price is higher than 45 at $\alpha =$
(a) 3.33 (b) 3.0 (c) 1.11 (d) 2.54

22. What is the table value for this test
(a) 1.96 (b) 2.33 (c) 1.64 (d) 2.58

23. What is the decision for the hypothesis
(a) Reserve judgment (b) Reject H_0 (c) Cannot reject H_0 (d) Accept H_0

24. The variance of the data set 4, 8, 12, 16, 20 is 40 then the variance of the data set 2, 4, 6, 8, 10 is -----
(a) 40 (b) 10 (c) 160 (d) 80

25. The variance of the data set 4, 6, 8, 10, 12, 14, 16, 18, 20 is 30 then the variance of the data set 4, 6, 8, 10, 12, 14, 16, 18 is -----
(a) 60 (b) 30 (c) 90 (d) 120

Use the following information to answer questions 26 to 28.

A random sample of 34 Shoprite customers spent a mean amount of 4.43 and standard deviation of 3.23

26. The calculated value of the statistic for testing the claim that the mean amount spent is less than 5 at level of significance is
(a) 1.03 (b) 1.11 (c) -1.03 (d) -1.11

27. What is the table value for this test
(a) 1.96 (b) 2.33 (c) 1.65 (d) 2.58

28. What is the decision for the hypothesis
(a) Reserve judgment (b) Reject H_0 (c) Accept H_0 (d) Cannot accept H_0

29. Tabular, graphical and numerical methods used to summarize data set is called ----- statistics
(a) Data presentation (b) Inferential (c) Descriptive (d) bar chart

30. In measuring sample, we normally summarize the sample ----- to be able to generalize the population
(a) Variable (b) parameter (c) statistic (d) survey

Use the following information to answer questions 31 to 33.

A random sample of 28 FUPRE students read a mean of 1.2 books with a standard deviation of 2.14 books a month.

31. The calculated value of the statistic for testing the claim that the mean number of books read per month is less than 2 at 0.05 level of significance is
(a) -0.80 (b) 0.80 (c) 1.98 (d) -1.98

32. What is the table value for this test
(a) 1.96 (b) 1.65 (c) 2.252 (d) 1.703

33. What is the decision for the hypothesis
(a) Reserve judgment (b) Accept H_0 (c) Cannot reject H_0 (d) Reject H_0

34. When equal interval is used in measuring population or sample, it is better to use -----
(a) ordinal scale (b) ratio scale (c) nominal scale (d) Interval scale

35. If $\text{Var}(X) = 2$, then $\text{Var}(X + C)$ where $C = 2$ is ----- (a) 6 (b) 4 (c) 8 (d) 2

Use the information on colour preference by gender in the table to answer questions 36 to 40.

Gender	Colour			
	Bright	Light	Dull	Other
Male	15	18	35	15
Female	35	12	10	10

50 30 45 25 150

60. A frequency distribution is constructed using the interval 1 - 5, 6 - 10, 11 - 15, etc. What is the class boundary of the third class?
 (a) 10.5 - 15.5 (b) 5.5 - 10.5 (c) 15.5 - 20.5 (d) 20.5 - 25.5

The table below is the frequency distribution of marks scored in Statistics. Use it to answer questions 61 - 65.

Marks	4	5	6	7	8
Frequency	5	9	8	6	2

61. What is the modal mark?
 (a) 9 (b) 5 (c) 8 (d) 6
62. Find the median mark?
 (a) 9 (b) 6 (c) 5 (d) 8
63. Find the mean mark?
 (a) 9.1 (b) 5.7 (c) 5.5 (d) 6.1
64. Find the variance of the distribution
 (a) 1.16 (b) 1.24 (c) 1.18 (d) 1.39
65. What is the standard deviation of the distribution?
 (a) 1.37 (b) 1.16 (c) 1.34 (d) 1.18

The table below shows test scores of a group of students. Use the table to answer question 66 to 70.

Class	Frequency
1 - 5	6
6 - 10	5
11 - 15	8
16 - 20	12
21 - 25	9
26 - 30	7

66. Find $\bar{x}$ if the mean of the frequency distribution is 15.8
 (a) 12.00 (b) 15.00 (c) 13.00 (d) 14.00
67. What is the median of the frequency distribution?
 (a) 10.92 (b) 15.80 (c) 15.92 (d) 14.9
68. What is the mode of the frequency distribution?
 (a) 13.00 (b) 14.00 (c) 14.90 (d) 15.50
69. What is the variance of the frequency distribution?
 (a) 7.37 (b) 7.29 (c) 54.24 (d) 53.16
70. Find the standard deviation of the frequency distribution
 (a) 7.29 (b) 54.24 (c) 7.37 (d) 53.16

Best of Luck.

Examiners: Dr. A. M. Okedeye, S. O. Ogundele, J. T. Eghwerido.

Paper 2 of 2 — MTH 112 FUNCTION AND GEOMETRIC

Level 100 • First Semester • Handout

DEPARTMENT OF MATHEMATICS AND STATISTICS

COURSE CODE: MTH 112

COURSE TITLE: FUNCTIONS AND GEOMETRY

LECTURES NOTES FOR NDS1, NDCS1 & DFST

PRESENTED BY: MR. D. M. FWACHABE

PART A: FUNCTION

FUNCTIONS

A function of x is an expression which depends on x . examples of functions of x are $x \log x$, $x^2 + 5x - 4$ and $\sin 35x$. A function of x is conventionally written $f(x)$. The brackets do not indicate multiplication but mean simply that the function depends upon x and not some others variable.

Consider the following exercises:

Given that: $x \in \{1, 2, 3, 4, 5\}$ find the corresponding set of values of y , where y is given by the rule:

(a) $y = x^2$

1

(b) $y =$

□

(c) $y = 5 - x$

THE ANSWERS ARE

(a) $x \begin{matrix} 1 & 2 & 3 & 4 & 5 \end{matrix}$ (b) $x \begin{matrix} 1 & 2 & 3 & 4 & 5 \end{matrix}$

$y \begin{matrix} 1 & 4 & 9 & 16 & 25 \end{matrix}$ $y \begin{matrix} 1 & 1 & 1 & 1 & 1 \end{matrix}$

$\begin{matrix} 2 & 3 & 4 & 5 \end{matrix}$

(c) $x \begin{matrix} 1 & 2 & 3 & 4 & 5 \end{matrix}$

$y \begin{matrix} 4 & 3 & 2 & 1 & 0 \end{matrix}$

$\begin{matrix} 1 & 1 & 1 & 1 \end{matrix}$

(a) $\{1, 4, 9, 16, 25\}$ (b) $\{1, , , , \}$ (c) $\{4, 3, 2, 1, 0\}$

$\begin{matrix} 2 & 3 & 4 & 5 \end{matrix}$

In the above examples, a set of values is given for the variable (x). Then rule is given and this is applied to the given set of numbers, to produce a set of values of the other variables (y). The variable (x) for which the values are given is called the independent variable and the set if values of the independent variable is called domain i.e $\{1, 2, 3, 4, 5\}$. The rule which is applied to the independent variable is

1

called the function and the variable which is produced by the rule is called the dependent variable.

The set of values of the dependent variable (y) is called range. In example (a) above, the range is the set $\{1, 4, 9, 16, 25\}$. The member of the range which corresponds to a certain member of the domain is called the image of that member, example in (a) above 4 is the image of 2, 9 is the image of 3, 16 is the image of 4 and 25 is the image of 5.

Note that there is no objection to having two distinct members of the domain with the same image.

THE FUNCTION NOTATION

Let us consider, two functions, both having set of real numbers as the domain. Let one of the functions squares each member of the domain and the other doubles each member of the domain.

If we write $f(x)$ to represent the image of x under the function f , then

1. The first function is $f(x) = x^2$
2. The first function is $g(x) = 2x$

The usual letters to use for this purpose are f, g, h and their corresponding capital letters.

Example 1:

- i. $f(2) = 2^2 = 4$ ii. $f(5) = 5^2 = 25$
- ii. $g(2) = 2 \times 2 = 4$ iv. $g(5) = 2 \times 5 = 10$

2

Example 2: The domain of the function $g(x) = 5x + 1$ is $\{0, 1, 2, 3, 4, 5\}$. Find its range.

$$g(x) = 5x + 1$$

$$g(0) = 5(0) + 1 = 1, g(1) = 5(1) + 1 = 6, g(2) = 5(2) + 1 = 11,$$

$$g(3) = 5(3) + 1 = 16, g(4) = 5(4) + 1 = 21, g(5) = 5(5) + 1 = 26$$

$$\text{It's range} = \{1, 6, 11, 16, 21, 26\}$$

Example 3: The range of the function $g(x) = x^2 + 1$ is $\{1, 2, 5, 10, 17, 26\}$.

Find its domain

$$g(x) = x^2 + 1$$

$$\therefore x^2 + 1 = 1 \quad x^2 + 1 = 2 \quad x^2 + 1 = 5 \quad x^2 + 1 = 10$$

$$x^2 = 0 \quad x^2 = 1 \quad x^2 = 4 \quad x^2 = 9$$

$$x = 0 \quad x = \sqrt{1} \quad x = \sqrt{4} \quad x = \sqrt{9}$$

$$x = \pm 1 \quad x = \pm 2 \quad x = \pm 3$$

$$x^2 + 1 = 17 \quad x^2 + 1 = 26$$

$$x^2 = 16 \quad x^2 = 25$$

$$x = \sqrt{16} \quad x = \sqrt{25}$$

$$x = \pm 4 \quad x = \pm 5$$

the domain is $\{0, \pm 1, \pm 2, \pm 3, \pm 4, \pm 5\}$ i.e. $-5 \leq x \leq 5$

3

Example 4:

Given that $f(x) = x^3 + 1$, find the values of

3

(a) $f(0)$ (b) $f(5)$ (c) $f(\quad)$ (d) $f(-2)$

4

Solution

(a) $f(x) = x^3 + 1 \therefore f(0) = 0^3 + 1 = 1$

(b) $f(5) = 5^3 + 1 = 5 \times 5 \times 5 + 1 = 125 + 1 = 126$

$3 \times 3 \times 3 + 1 = 27 + 1 = 28$

(c) $f(\quad) = f(\quad)^3 + 1 = x \times x \times x + 1 = \quad + 1 = 1$

$4 \times 4 \times 4 + 1 = 64 + 1 = 65$

(d) $f(-2) = (-2)^3 + 1 = (-2) \times (-2) \times (-2) + 1 = -8 + 1 = -7$

WORKED EXAMPLES

1

1. the domain of the function $f(x) =$ is

$(1 + \sqrt{x})^2$

find its range

$1 \leq x \leq 1$

$f(x) = \quad, f(0) = \quad, f(1) = \quad, f(x) = \quad$

$1 + \sqrt{x} \geq 1 \therefore 1 + 0 \leq 1 + \sqrt{x} \leq 1 + 1 \therefore 1 \leq f(x) \leq 2$

$\therefore \text{the range is } \{y: y \in \mathbb{R}, 1 \leq y \leq 2\}$

2. Given that $f(x) = x^2 - x$, find the values of $f(10)$, $f(-3)$

$f(10) = 10^2 - 10 = 100 - 10 = 90$

$f(-3) = (-3)^2 - (-3) = 9 + 3 = 12$

EXERCISE 1

1. What is the range of the function $f: \mathbb{R} \rightarrow \mathbb{R}$ having as its domain $-3 \leq x \leq$

4.

Answer: $-3 \leq x \leq 11$ or $\{-3, -1, 1, 3, 5, 9, 11\}$

4

2. The domain of the function $f(x) = x^2 + 1$ is $\mathbb{R}$. find its range.

Answer: $\{y: y \in \mathbb{R}, y \geq 1\}$ or $\{1, 2, 3, 4, \dots\}$

3. Given that $f(x) = x^3$, find

1

(a) $f(2)$ (b) $f(-10)$ (c) $f(\quad)$ (d) $f(5)$

2

1

Answer: (a) 8 (b) -1000 (c) $\frac{1}{125}$ (d) 125

8

4. Given that $g(x) = x^3 + 1$, find the values of

3

(a) $g(0)$ (b) $g(5)$ (c) $g(\)$ (d) $g(-2)$

4

FUNCTIONS/MAPPINGS

(a) 'Into' Mapping (Function): The mapping is such that every element of the set X has its image in set Y (no member is left out without its image). This type of mapping is called mapping of X into Y .

From the above fig. 1, for example, under the rule 'f' 'a' correspond to 1, 'b' to 2 and also 'c' to 2; but 3 and 4 are no images of any element of X . This is the mapping of X into Y . Note that fig. 2 does not represent a function (mapping) of A and B because there is nothing assigned to the element $\square \in \square$.

5

(b) 'Onto' Mapping (Function):

In the situation where every element of Y is an image of some elements of X , then this type of mapping is called mapping of X onto Y . for example;

We can also say that f is onto if and only if the range of f is the entire co-domain.

(c) One - to - One Mapping (Function):

A function $f: X \rightarrow Y$ is said to be one - to - one if different elements in X have distinct image in Y .

For example: Consider the function $f: X \rightarrow Y$ defined by the diagram below. Does it represent a one - to - one mapping?

6

Yes, f is one - to - one mapping since no element of B is the image of more than one element of A .

Another example of one - to - one mapping is below

COMPOSITE FUNCTIONS

Let f and g represent the functions $f(x) = x^2$ and $g(x) = x + 5$. The domain of both function is (set of real numbers). Therefore, $f(3) = 3^2 = 9$ and $g(x) = 9 + 5 = 14$.

Notice that we applied the first function f and then functioning g to obtain 14. This could be written as $g[f(3)] = g(9) = 9 + 5 = 14$, it usually abbreviated to

$$gf(3) = 14$$

Example 1:

Given that $f: X \rightarrow X + 1$ and that $g: X \rightarrow x^2$, express the composite functions fg and gf in their simplest possible forms.

7

Solution

$$f(x) = 5x+1 \text{ and } g(x) = x^2$$

$$fg = f[g(x)] = f(x^2) = 5x^2 + 1$$

$$gf = g[f(x)] = g(5x + 1) = (5x + 1)^2$$

Example 2:

The function f and g are defined by: $f: \mathbb{R} \rightarrow \mathbb{R}$ and that $g: \mathbb{R} \rightarrow x + 3$, where $x \in \mathbb{R}$ define in similar manner (a) fg (b) gf

(G.C.E O/L University of London, Jan.1983, Paper 1)

Answer:

$$(a) fg(x) = f[g(x)] = f(x + 3) = x + 3 + 3 = x + 6$$

$$\therefore fg(x): \mathbb{R} \rightarrow x + 6$$

$$(b) gf(x) = g[f(x)] = g(2x) = 2x + 3$$

$$\therefore gf(x): \mathbb{R} \rightarrow 2x + 3$$

Example 3:

Given that $f: \mathbb{R} \rightarrow 5x + 1$ and that $g: \mathbb{R} \rightarrow x^2$, express the composite functions fg and gf in their simplest possible forms.

Find the values of $fg(2)$, $gf(2)$, $fg(3)$, $gf(4)$

Solution

$$\text{Given } f: \mathbb{R} \rightarrow 5x + 1 \text{ and } g: \mathbb{R} \rightarrow x^2$$

$$fg = f[g(x)] = f(x^2) = 5x^2 + 1$$

$$\begin{aligned} gf &= g[f(x)] = g(5x + 1) = (5x + 1)^2 = (5x + 1)(5x + 1) \\ &= 25x^2 + 10x + 1 \end{aligned}$$

8

$$fg(2) = 5(2)^2 + 1 = 5 \times 4 + 1 = 20 + 1 = 21$$

$$gf(2) = (5 \times 2 + 1)^2 = (10 + 1)^2 = 11^2 = 11 \times 11 = 121$$

$$fg(3) = 5(3)^2 + 1 = 5 \times 9 + 1 = 45 + 1 = 46$$

$$gf(4) = (5 \times 4 + 1)^2 = (20 + 1)^2 = 21^2 = 21 \times 21 = 441$$

Example 4:

Find the rule for the product mappings

$$(a) f^2 = fg \quad (b) f^2 = gf \quad (c) fg \quad (d) gf$$

$$\text{If } f: \mathbb{R} \rightarrow 2x - 3 \text{ and } g: \mathbb{R} \rightarrow x^2 - 2x + 3$$

(e) Check whether fg and gf are equal or not.

Solution

$$(a) \text{ Given that } f: \mathbb{R} \rightarrow 2x - 3, g: \mathbb{R} \rightarrow x^2 - 2x + 3$$

$$\begin{aligned} f^2(x) &= f[g(x)] = f(x^2 - 2x + 3) = 2(x^2 - 2x + 3) - 3 \\ &= 4x^2 - 6x + 3 \end{aligned}$$

$$= 4x^2 - 9$$

$$\begin{aligned} (b) f^2(x) &= g[f(x)] = g(2x - 3) = (2x - 3)^2 - 2(2x - 3) + 3 \\ &= x^4 - 4x^3 + 10x^2 - 12x + 9 - 2x^2 + 4x - 6 + 3 \end{aligned}$$

$$= x^4 - 4x^3 + 8x^2 - 8x + 6$$

$$(c) \quad \square\square(x) = \square[\square(\square)] = \square(x^2 - 2x + 3) = 2(x^2 - 2x + 3) - 3$$

$$= 2x^2 - 4x + 6 - 3$$

$$= 2x^2 - 4x + 3$$

$$(d) \quad \square\square(x) = \square[\square(\square)] = \square(2x - 3)^2 - 2(2x - 3) + 3$$

$$9$$

$$= 4x^2 - 12x + 9 - 4x + 6 + 3$$

$$= 4x^2 - 16x + 18$$

(d) Since $\square\square(x) = 2x^2 - 4x + 3$ and $\square\square(x) = 4x^2 - 16x + 18$, therefore, $\square\square \neq \square\square$. This shows that composite of Mapping is not commutative in general.

Example 5:

For each pair of functions $\square$ and $\square$ solve the equation $\square\square(\square) = \square\square(\square)$, for values of $\square$ where the compositions $\square\square$ and $\square\square$ coincide.

$$(a) \quad \square: \square \rightarrow 2\square - 1 \quad \square: \square \rightarrow x^2 - x + 2$$

$$(b) \quad \square: \square \rightarrow x^2 + 1 \quad \square: \square \rightarrow \sqrt{3\square - 2}$$

Solution

$$(a) \quad \square\square(x) = \square[\square(\square)] = \square(x^2 - x + 2) = 2(x^2 - x + 2) - 1$$

$$= 2x^2 - 2x + 4 - 1$$

$$= 2x^2 - 2x + 3$$

$$(b) \quad \square\square(x) = \square[\square(\square)] = \square(2x - 1) = (2x - 1)^2 - (2x - 1) + 2$$

$$= (-2x - 1)(2x - 1) - (2x - 1) + 2$$

$$= 4x^2 - 6x + 4$$

Since $\square\square(x) = \square\square(x)$, therefore

$$2x^2 - 2x + 3 = 4x^2 - 6x + 4$$

$$\therefore 4x^2 - 2x^2 - 6x + 2x + 4 - 3$$

$$= 2x^2 - 4x + 1 = 0$$

$$10$$

The L.H.S cannot be factorized. We can use the "almighty" formula for the solutions of the quadratic equation.

$$-\square \pm \sqrt{\square^2 - 4\square\square}$$

$$\therefore \square =$$

$$2\square$$

Where $a = 2$, $b = -4$, $c = 1$

$$-(-4) \pm \sqrt{(-4)^2 - 4(2)(1)} \quad 4 \pm \sqrt{8}$$

$$\square =$$

$$2(2) \quad 2(2)$$

$$4 \pm \sqrt{4} \times 2 \quad 4 \pm 2\sqrt{2} \quad \sqrt{2}$$

$$= = = 1 \pm$$

4 4 2

1

$$= 1 \pm 2\sqrt{2}$$

(b) $\square: \square \rightarrow x^2 + 1$ $\square: \square \rightarrow \sqrt{3\square - 2}$

$$\square\square(\square) = \square[\square(\square)] = \square(\sqrt{3}\square - 2) = (\sqrt{3}\square - 2)^2 + 1 = 3\square - 2 + 1 = 3\square - 1$$

$$\square\square(\square) = \square[\square(\square)] = \square(x^2 + 1) = \sqrt{3}(\square^2 + 1) - 2 = \sqrt{3}\square^2 + 1$$

But $\varphi(\varphi) = \varphi(\varphi) \therefore$

$$3n - 1 = \sqrt{(3n^2 + 1)}$$

$$(3x - 1)^2 = 3x^2 + 1$$

$$9x^2 - 6x + 1 = 3x^2 + 1$$

$$6 \times 2 - 6 = 0$$

11

$$\square(6\square - 6) = 0 \implies 6\square(\square - 1) = 0$$

$$x = 0, x = 1 \therefore 6x = 0 \implies x - 1 = 0$$

$$x=0 \quad x=1$$

EXERCISE B

1. Is the mapping one - to - one or onto? Where $\mathbb{A} = \{-3, -2, -1, 1, 2\}$, $\mathbb{B} = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25\}$ and $f: \mathbb{A} \rightarrow \mathbb{B}$ + assigns to every number in A, its square.

2. Determine whether or not the mapping is one - to - one or onto

$$\square: \square \times \square, \square(\square, \square) \rightarrow (\square - 4\square, -2\square).$$

3. If $\square(\square) = x^2 - 4x - 5$, find the values of the following.

(a) $\square(2)$ (b) $\square(1)$ (c) $\square(0)$ (d) $\square(\square + 1)$ (e) $\square(-5)$

4. The domain of the function $\square(\square) = 3\square + 1$ is $\{0, 1, 2, 3, 4\}$. Find its range.

5. The domain of the function $\varphi(x) = x^2 + 2$ is $\mathbb{R}$. find its range.

6. (a) If $\square(\square) = \square 2$, find the value of the following:

(i) $\square(3)$ (ii) $\square(3.1)$ (iii) $\square(3.001)$

$$\varphi(3.001) - \varphi(3)$$

(b) Also find the value of

0.001

7. If $\square: \square \rightarrow \square + 1$ what is the function $\square\square$? Find the values of the following

(a) $\sin(2)$ (b) $\sin(-1)$ (c) $\sin(4)$ (d) $\sin(0)$

12

8. Given that $\square(\square) = x^2$ express as simple as possible

$$\Pi(2+\Pi)-\Pi(2)$$

(a) $\square(2 + \square)$ (b) $\square$, $(\square \neq 0)$

$\square$

9. If $\square: \square \rightarrow 2\square$ and $\square: \square \rightarrow \square\square\square 0$, find the values of

(a) $\square(10)$ (b) $\square(10)$ (c) $\square\square(10)$ (d) $\square\square(10)$ (e) $\square\square(15)$ (d) $\square\square(15)$

10. Given that $\square: \square \rightarrow 3\square + 1$ and $\square: \square \rightarrow \square 2$, determine the formulae for the composite functions $\square\square$ and $\square\square$ in their simplest possible forms.

11. If $\square(\square) = 2\square 2$, find expressions for:

$\square(\square+h) - \square(\square)$

(a) $\square(\square + h)$ (b) $\square(\square + h) - \square(\square)$ (c) $\square$, $(h \neq 0)$

h

12. If $\square: \square \rightarrow \square - 1$ and $\square: \square \rightarrow 2\square$, express the composite function $\square\square$ and $\square\square$ in their simplest possible forms. What are the values of

$\square\square(2)$, $\square\square(2)$, $\square\square(-1)$ and $\square\square(-1)$? Illustrate the composite

functions using arrow diagrams with three parallel lines taking as domain in each case the set $\{-1, 0, 2, 5\}$

13. If $\square: \square \rightarrow 3 - 4\square$ and $\square: \square \rightarrow 5\square - 1$. Solve the equation $\square\square(\square) = \square\square(\square)$

14. For each pair of functions $\square$ and $\square$ solve the equation $\square\square(\square) = \square\square(\square)$, for the values $\square$ where the compositions $\square\square$ and $\square\square$ coincide.

(a) $\square: \square \rightarrow 1 - 3\square$ $\square: \square \rightarrow 2 - \square 2$

13

(b) $\square: \square \rightarrow \sqrt{2\square - 3}$ $\square: \square \rightarrow 3\square 2 + 1$

(c) $\square: \square \rightarrow \square 2 + 1$ $\square: \square \rightarrow \sqrt{3\square - 2}$

EXERCISE B (ANSWERS)

1. (i) Not one - to - one mapping

(ii) Not onto mapping

2. (i) One - to - one mapping

(iii) Onto mapping

3. (a) 7 (b) 0 (c) -5 (d) $a(a+6)$ (e) 0

4. $\{1, 4, 7, 10, 13\}$

5. $\{3, 6, 11, 18, 27, \dots \dots \}$

6a. (i) 9 (ii) 9.61 (iii) 9.006001

6b. 6.001

7. $\square\square: \square \rightarrow \square + 1$ (a) 4 (b) 1 (c) 6 (d) 2

8. (a) $4 + 4\square + \square 2$ (b) $4 + \square$

9. (a) 20 (b) $\square\square\square 100$ (c) $2 \square\square\square 100$ (d) $\square\square\square 200$ (e) $2 \square\square\square 150$

$$f: \mathbb{R} \rightarrow 2\mathbb{R}$$

$$(e) f(300) = ?$$

$$f: \mathbb{R} \rightarrow 2\mathbb{R}$$

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$$10. f: \mathbb{R} \rightarrow 3x^2 + 1, g: \mathbb{R} \rightarrow (3x + 1)^2$$

$$11. (a) 2x^2 + 4xh + 2h^2 (b) 4xh + 2h^2 (c) 4xh + 2h$$

$$12. f: \mathbb{R} \rightarrow 2x - 1, g: \mathbb{R} \rightarrow 2(x - 1); 3, 2, -3, -4$$

1 8 345

$$13. \text{No solution } 14 (a) x = 1 (b) x = \pm\sqrt{}$$

2 5 30

1.7 INVERSE FUNCTIONS

In a situation where a function f maps a member x of set A onto unique member y of set B and a function g maps y of B onto x of A . For g to exist, the range of f in B must be entire set B , if not, certain members of B would not have image in A under f . Again, for g to exist, the relation between x and y under f must be of the kind called one - to - one correspondence which is shown in the arrow diagram below.

When the above situation occurs then g is called the inverse function to f written $g = f^{-1}$. Many simple inverse functions can be written at sight, for example:

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$$(a) f(x) = x + 1 \quad f^{-1}(x) = x - 1$$

$$f$$

$$(b) f(x) = x^2 - 1 \quad f^{-1}(x) = \pm\sqrt{x+1}$$

$$f$$

$$(c) f(x) = x^2 - 1 \quad f^{-1}(x) = \sqrt{x+1}$$

$$(d) f(x) = x - 1 \quad f^{-1}(x) = x + 1$$

1 1

$$(e) f(x) = x - 1 \quad f^{-1}(x) = x + 1$$

$$f$$

$$7 - 3x$$

Example 10: Find the inverse of $f(x) =$

10

Solution

$$7 - 3x$$

$$f \rightarrow y$$

10

$$7 - 3x$$

$$\text{So } y = 10, 7 - 3x = 10$$

10

$$7-10$$

$$3$$

$$7-10$$

$$= \leftarrow : -1$$

$$3$$

$$7-10$$

Thus using as the starting value, $-1 : \rightarrow$

$$3$$

Example 11: If $\rightarrow 3$ and $\rightarrow + 2$. Find

(a) (b) -1 (c) -1 (d) -1 (e) $-1 -1$

$$16$$

Solution

$$(a) (\rightarrow) = \rightarrow(\rightarrow) = \rightarrow(+2) = 3(+2) = 3 + 6$$

$$(b) (\rightarrow)-1, \rightarrow \rightarrow 3 + 6 =$$

$$-6$$

$$\text{So } 3 + 6 = \therefore =$$

$$3$$

$$-6$$

$$= \leftarrow : (\rightarrow)-1$$

$$3$$

$$-6$$

$$\therefore (\rightarrow)-1 : \rightarrow$$

$$3$$

$$(c) -1, \rightarrow \rightarrow 3 =$$

$$-$$

$$\text{So } 3 = \therefore =$$

$$3$$

$$-$$

$$= \leftarrow : -1$$

$$3$$

$$-$$

$$\therefore -1 : \rightarrow$$

$$3$$

$$(d) -1, \rightarrow + 2 =$$

$$\text{So } + 2 = \therefore = -2$$

$$-2 = \leftarrow : -1$$

$$-6$$

$$(e) (-1 -1) = -1 [-1 (-)] = -1 () = -2 =$$

3 3 3

17

Example 12: If $f: \mathbb{R} \rightarrow 5 - \mathbb{R}$, find f^{-1}

Solution

$$f: \mathbb{R} \rightarrow 5 - \mathbb{R} = \mathbb{R}$$

$$\text{So } 5 - \mathbb{R} = \mathbb{R} \therefore 5 - \mathbb{R} = \mathbb{R}$$

$$5 - \mathbb{R} = \mathbb{R} \leftarrow f: \mathbb{R} \rightarrow 5 - \mathbb{R}$$

$$\mathbb{R} \rightarrow 5 - \mathbb{R} : \mathbb{R} \rightarrow 5 - \mathbb{R}$$

Note that $\mathbb{R} \rightarrow 5 - \mathbb{R}$ is the same function as f . This function is an example of self-inverse.

EXERCISE C

1. Given that $f(x) = 7x + 3$, find the values of

(a) $f^{-1}(45)$ (b) $f^{-1}(-11)$ (c) $f^{-1}(-18)$

2. Find the inverses of functions for each of the following

$$f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = 3x - 1$$

(a) $f: \mathbb{R} \rightarrow \mathbb{R}$ (b) $f: \mathbb{R} \rightarrow \mathbb{R}$ (c) $f: \mathbb{R} \rightarrow \mathbb{R}$

$$f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = 2x + 4$$

3. Find the universes of the following functions

$$f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = 3x - 1$$

(a) $f: \mathbb{R} \rightarrow \mathbb{R} (x \neq 4)$ (b) $f: \mathbb{R} \rightarrow \mathbb{R} (x \neq 2)$ (c) $f: \mathbb{R} \rightarrow \mathbb{R} (x \neq -3)$

$$f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = 4 - 2x + 1$$

$$f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = 2x + 2$$

4. Show that $f(x) = \frac{1}{x}$ is a self inverse function.

$$f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = \frac{1}{x}$$

5. If $f: \mathbb{R} \rightarrow 3\mathbb{R}$ and $g: \mathbb{R} \rightarrow \mathbb{R} \setminus \{0\}$, find formulae for f^{-1} and g^{-1}

$$f: \mathbb{R} \rightarrow 3\mathbb{R}, f(x) = 3x$$

6. Shows that the function $f(x) = \frac{1}{x-2} (x \neq 2)$, is the inverse of the function $g(x) = \frac{1}{2-x} (x \neq 2)$.

$$f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = \frac{1}{x-2}$$

$$g: \mathbb{R} \rightarrow \mathbb{R}, g(x) = \frac{1}{2-x}$$

($x \neq 2$).

$$f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = \frac{1}{x-2}$$

$$g: \mathbb{R} \rightarrow \mathbb{R}, g(x) = \frac{1}{2-x}$$

$$f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = \frac{1}{x-2}$$

7. The function $f(x) = \frac{1}{x}$ and $g(x) = \frac{1}{x} (x \neq 0)$, determine which is self-inverse

$$f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = \frac{1}{x}$$

function.

ANSWERS EXERCISE C

1. (a) 6 (b) -2 (c) -3

$$f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = 3x - 1$$

2. (a) $2\pi - 1$ (b) (c)

3 π

$4\pi - 3$ $1 - \pi$ 2π

3. (a) (b) (c)

π 2π $2 - \pi$

1 π 0

4. $\sin 0$, $\cos 0$

3 3

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DEPARTMENT OF MATHEMATICS AND STATISTICS

COURSE CODE: MTH 112

COURSE TITLE: FUNCTIONS AND GEOMETRY

LECTURES NOTES FOR NDS1, NDCS & DFST

PRESENTED BY: MR. D. M. FWACHABE

PART B: TRIGONOMETRIC FUNCTIONS

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TRIGONOMETRIC FUNCTIONS

In a right - angled triangle ABC, with base angle 0 , the following ratios exist.

$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$

1) $\sin \theta = \frac{y}{r}$, $-1 \leq \sin \theta \leq 1$: (opposite side is always less than hypotenuse.)

$\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}}$

$\tan \theta = \frac{\text{opposite}}{\text{adjacent}}$

2) $\cos \theta = \frac{x}{r}$, $-1 \leq \cos \theta \leq 1$: (adjacent side is always less than hypotenuse.)

$\sec \theta = \frac{1}{\cos \theta}$

$\csc \theta = \frac{1}{\sin \theta}$

3) $\tan \theta = \frac{y}{x}$, $-\infty \leq \tan \theta \leq \infty$

$\cot \theta = \frac{1}{\tan \theta}$

$\sec \theta = \frac{1}{\cos \theta}$

4) $\csc \theta = \frac{1}{\sin \theta}$, $-\infty < \csc \theta < \infty$

$\sec \theta = \frac{1}{\cos \theta}$

$\csc \theta = \frac{1}{\sin \theta}$

5) $\sec \theta = \frac{1}{\cos \theta}$, $|\sec \theta| \geq 1$

$\csc \theta = \frac{1}{\sin \theta}$

$\sec \theta = \frac{1}{\cos \theta}$

6) $\csc \theta = \frac{1}{\sin \theta}$, $|\csc \theta| \geq 1$

$\sec \theta = \frac{1}{\cos \theta}$

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Example 1: Given a right - angled triangle ABC where the side AB=4, AC=5 and

CB=3. Find the values

(a) $\sin \theta$ (b) $\cos \theta$ (c) $\tan \theta$ (d) $\sec \theta$ (e) $\csc \theta$

(f) $\cot \theta$

Solution

3 3 4 5 5

(a) $\sin \theta = \frac{3}{5}$ (b) $\cos \theta = \frac{4}{5}$ (c) $\tan \theta = \frac{3}{4}$ (d) $\sec \theta = \frac{5}{4}$ (e) $\csc \theta = \frac{5}{3}$

5 5 3 3 4

3

(f) $\cot \theta = \frac{4}{3}$

4

EXAMPLE 2:

In a right-angled triangle ABC if the length of AB=12, BC =5, write down the values of (i) $\tan \theta$ (ii) $\cot \theta$ (iii) $\sec \theta$ (iv) $\sin \theta$

Using Pythagoras theorem to find the length of AC

$$AC = \sqrt{12^2 + 5^2} = \sqrt{144 + 25} = \sqrt{169} = 13$$

$\therefore$ the length of AC = 13 as illustrated in the diagram below

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5 5 13 12

(i) $\tan \theta = \frac{5}{12}$ (ii) $\cot \theta = \frac{12}{5}$ (iii) $\sec \theta = \frac{13}{12}$ (iv) $\sin \theta = \frac{5}{13}$

12 12 12 13

EXAMPLE 3:

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Given that $\cos \theta = \frac{21}{29}$, find (i) $\tan \theta$ (ii) $\csc \theta$

29

By Pythagoras theorem, $BC = \sqrt{29^2 - 21^2} = \sqrt{29^2 - 21^2}$

$$BC = \sqrt{841 - 441} = \sqrt{400}$$

BC = 20

20 29

(i) $\tan \theta = \frac{20}{21}$ (ii) $\csc \theta = \frac{29}{20}$

21 20

23

RATIO OF ANGLES OF ANY MAGNITUDE

1. An angle greater than 360° can be reduced to less than 360° by subtracting from it, multiples of 360°. Example: $\theta = 8900^\circ - 7200^\circ = 1700^\circ$

2. An angle less than 360° is either greater than 180° or less than 180°

Example: (i) $1700^\circ = (1800^\circ - 100^\circ)$ (ii) $2250^\circ = (1800^\circ + 450^\circ)$

(iii) $3000^\circ = (3600^\circ - 600^\circ)$

Let us consider the table below

1st Quadrant, $0 \leq \theta \leq 90^\circ$. All ratios are positive

2nd Quadrant, $90^\circ < \theta \leq 180^\circ$. Only $\sin \theta$ and $\csc \theta$ are positive the remaining four ratios are negative.

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3rd Quadrant, $180^\circ < \theta \leq 270^\circ$. Only $\tan \theta$ and $\sec \theta$ are positive the remaining four ratios are negative.

4th Quadrant, $270^\circ < \theta \leq 360^\circ$. Only $\cos \theta$ and $\sec \theta$ are positive the remaining four ratios are negative.

To find the trigonometric ratios of the angle of any magnitude

(a) First reduce the angle as a combination of $180^\circ \pm \theta$ or $(360^\circ - \theta)$ then

(b) Assign it a positive or negative sign according to the quadrant it becomes.

For example:

$$\sin 170^\circ = + \sin(180^\circ - 10^\circ) = + \sin 10^\circ$$

$$\cos 170^\circ = - \cos(180^\circ - 10^\circ) = - \cos 10^\circ$$

$$\sin 225^\circ = + \sin(180^\circ + 45^\circ) = + \sin 45^\circ$$

$$\cos 225^\circ = - \cos(180^\circ + 45^\circ) = - \cos 45^\circ$$

$$\sin 300^\circ = - \sin(360^\circ - 60^\circ) = - \sin 60^\circ$$

$$\cos 300^\circ = + \cos(360^\circ - 60^\circ) = + \cos 60^\circ$$

We should note that for angles $-\theta$, $180^\circ \pm \theta$, $360^\circ \pm \theta$, sines remain sines, cosines remain cosines and tangents remain tangents. For $90^\circ \pm \theta$, $270^\circ \pm \theta$, sines become cosines, cosines become sines and tangents becomes cotangents. For example: (i) $\sin(270^\circ - \theta) = -\cos \theta$ (ii) $\sin(360^\circ + \theta) = \sin \theta$

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MEASUREMENT OF ROTATION

When a line OP is pivoted at O as shown above and rotates from its initial position OP₀ to a new position OP₁, the angle P₀ O P₁ is a measure of the rotation of OP. The angle θ is usually measured in one of two units, namely; degrees or radians.

THE DEGREE

The ancient Babylonian mathematicians divided one complete revolution into 360 equal parts because they thought that the solar year was 360 days, each part now being known as one degree (1°). Using the degree as the unit of rotation, half a revolution corresponds to 180° and a quarter of revolution, i.e a right angle, corresponds to 90°

26

THE RADIAN

A radian is the angle subtended at the centre of a circle by an arc equal to the radius.

Diagram illustrating the definition of a radian:

The number of radian in one complete revolution =

□□□□□□

Circumference of a circle of radius, $r = 2\pi r$

$2\pi r$

$\therefore$ The number radians in one revolution = 2π

□

i.e π □□□□□□ = □□□□ (□□□□□□)

2π

Half a revolution = □□□□□□ = □ □□□□□□□ = 1800 (□□□□□□)

2

□

One right angle = □□□□□□ = 900 (□□□□□□)

2

Changing Degrees In An Angle to Radians

□

Multiply the number of degrees by the factor

180

For example: Express the following degrees in radians, giving your answer in terms of π .

(i) 600 (ii) 300 (iii) 1200 (iv) 1350

Solution

27

□ □

(i) $600 = 60 \times \pi = \pi$ □□□□□□

180 3

□ □

(ii) $300 = 30 \times \pi = \pi$ □□□□□□

180 6

□ 4□

(iii) $1200 = 120 \times \pi = \pi$ □□□□□□

180 6

□ 3□

(iv) $1350 = 135 \times \pi = \pi$ □□□□□□

180 4

Changing Radians to Degrees

180

Multiply the number radians by

□

For example: without using tables, express the following angles in degrees.

$$7^\circ 5' 3''$$

(i) (ii) (iii) (iv)

$$6^\circ 3' 4''$$

Solution

$$7^\circ 5' 3''$$

$$(i) = x = 7 \times 300 = 2100$$

$$6^\circ 6''$$

$$5^\circ 5' 180$$

$$(ii) = x = 5 \times 600 = 3000$$

$$3^\circ 3''$$

$$3^\circ 3' 180$$

$$(iii) = x = 3 \times 450 = 1350$$

$$4^\circ 4''$$

$$3^\circ 3' 180$$

$$(iv) = x = 3 \times 900 = 2700$$

$$2^\circ 2''$$

$$28$$

EXERCISE

Without using tables, express the following angles in radians, giving your answer in terms of π .

$$1. 2700 \quad 2. 450 \quad 3. 2400 \quad 4. 1500 \quad 5. 200$$

$$6. 3000 \quad 7. 22.50 \quad 8. 800$$

Without using tables, express the following angles in degree:

$$5^\circ 11' 4''$$

(9) (10) (11) (12)

$$6^\circ 10' 6''$$

DERIVE TRIGONOMETRIC IDENTITIES

Using Pythagoras theorem in a right – angled as below we can deduce some fundamental trigonometric identities.

The above triangle is a right – angled triangle, using Pythagoras' theorem we have

$$a^2 + b^2 = c^2$$

$$a^2 + b^2 = c^2$$

But $\sin^2 \theta + \cos^2 \theta = 1$ Divide through by c^2 to obtain

$$\frac{a^2}{c^2} + \frac{b^2}{c^2} = \frac{c^2}{c^2}$$

$$29$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$+ = 1$$

$$\sin^2 \theta$$

$$\therefore \sin^2 \theta + \cos^2 \theta \equiv 1$$

We can also deduce two similar identities by dividing the above identity through by

$\sin^2 \theta$, i.e

$$\sin^2 \theta + \cos^2 \theta \equiv 1$$

$$+ \equiv$$

$$\sin^2 \theta + \cos^2 \theta \equiv \sin^2 \theta$$

$$\sin^2 \theta \equiv 1$$

$$1 + \equiv$$

$$\sin^2 \theta + \cos^2 \theta$$

$$\cos^2 \theta \equiv 1$$

But $\cos^2 \theta = 1 - \sin^2 \theta$ and $\sin^2 \theta = 1 - \cos^2 \theta$ therefore $1 + \sin^2 \theta \equiv \sin^2 \theta$

$$\cos^2 \theta \equiv 1$$

Again, dividing the original identity by $\cos^2 \theta$ i.e $\sin^2 \theta + \cos^2 \theta \equiv 1$ divide through

by $\cos^2 \theta$ we get,

$$\sin^2 \theta + \cos^2 \theta \equiv 1$$

$$+ \equiv$$

$$\sin^2 \theta + \cos^2 \theta \equiv \cos^2 \theta$$

$$\sin^2 \theta \equiv 1$$

$$+1 \equiv$$

$$\sin^2 \theta + \cos^2 \theta$$

$$30$$

$$\cos^2 \theta \equiv 1$$

But $\sin^2 \theta = 1 - \cos^2 \theta$ and $\cos^2 \theta = 1 - \sin^2 \theta$

$$\sin^2 \theta \equiv 1$$

Therefore, $\sin^2 \theta + 1 \equiv \cos^2 \theta$

We should note the following trigonometric identities

$$\sin^2 \theta$$

$$1. \sin^2 \theta =$$

$$\cos^2 \theta$$

$$\sin^2 \theta$$

$$2. +$$

$$\sin^2 \theta \cos^2 \theta$$

$$\sin^2 \theta$$

$$3. \sin^2 \theta =$$

$$\cos^2 \theta$$

$$\sin^2 \theta$$

$$4. \sin^2 \theta \cos^2 \theta =$$

$$\sin^2 \theta$$

$$5. \sin^2 \theta + \cos^2 \theta = 1$$

$$6. \sin^2 \theta + \tan^2 \theta = \sec^2 \theta$$

$$7. \sin^2 \theta + \cot^2 \theta = \csc^2 \theta$$

THE GENERAL ANGLE

We should note that the axis divide the plane into four quadrants (as shows in the diagram below) and angles are measured in anti – clockwise direction from the x – axis, the quadrants are numbered as in the shown diagram below

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TRIGONOMETRICAL RATIO OF SPECIAL ANGLES

The trigonometrical ratios of angles 300 , 450 , 600 are used in mechanics and other branches of mathematics, therefore, we now deduce their values in surd form.

□□□ Using the square, ABCD, drawn with sides of unit length, AC, is the diagonal.

□□□

□□ 1

□□□ 450 = =

□□ $\sqrt{2}$

□□ 1

□□□ 450 = =

□□ $\sqrt{2}$

32

□□ 1

□□□ 450 = = =1

□□ 1

□□□ Using an equilateral triangle, ABC, of sides 2 units in length as drawn below and below and side AD is perpendicular to BC, ACD = 600

□□□

□□ $\sqrt{3}$

□□□ 600 = =

□□ 2

□□ 1

□□□ 600 = =

□□ 2

□□ $\sqrt{3}$

□□□ 600 = = = $\sqrt{3}$

□□ 1

□□□ Using CA

^ D = 300 in the above diagram

□□ 1

$$\cos 300 = -\frac{1}{2}$$

$$\sin 2$$

$$33$$

$$\cos \sqrt{3}$$

$$\cos 300 = -\frac{1}{2}$$

$$\sin 2$$

$$\cos 1$$

$$\cos 300 = -\frac{1}{2}$$

$$\sin \sqrt{3}$$

From the diagram below, let us consider the arm $\cos 1$. As the arm approaches the line $\cos$, the value of the angle approaches 0° , the y - component of the arm vanishes when the angle is 0° and the x - component becomes the same length as the arm, i. e. equal to unity.

For $\cos$

$$\cos 0$$

$$\cos 0 = 1$$

$$\sin 1$$

$$\cos 1$$

$$\cos 0 = 1$$

$$\sin 1$$

$$0$$

$$\cos 0 = 1$$

$$1$$

$$34$$

Consider the arm $\sin 2$ which is approaching 0° when the angle is 90° , the x - component of the arm becomes 0, and the y - component becomes unity (referring to the above diagram)

$$\sin 1$$

$$\sin 90 = 1$$

$$\cos 1$$

$$\sin 0$$

$$\sin 90 = 1$$

$$\cos 1$$

$$\sin 1$$

$$\sin 90 = 1$$

$$\cos 0$$

Referring to the above diagram, the arm $\sin 3$, the y - component decreases to zero as the angle approaches to 180° , and the x - component approaches a value of

−1.

□ 0

□□□ 1800 = = = 0

1 1

□ −1

□□□ 1800 = = = −1

1 1

□ 0

□□□ 1800 = = = 0

□ −1

□□□□ Referring to the above diagram again, the arm V4, as the vector approaches the axis Oy1, the x - component, which is negative approaches zero and the y - component approaches a value of −1.

35

□ −1

□□□ 2700 = = = −1

1 1

□ 0

□□□ 2700 = = = 0

1 1

□ −1

□□□ 2700 = = = ∞

□ 0

∴ □□□ 2700 has no value

Summary:

Let us write the values of all these angles in table

□□ □□□ □□□ □□□ □□□□ □□□□ □□□□ □□□□

□□□ 0 1 1 $\sqrt{3}$ 1 $\sqrt{3}$ 1 1 0

2 $\sqrt{2}$ 2 2 $\sqrt{2}$ 2

□□□ 1 $\sqrt{3}$ 1 1 0 −1 −1 $-\sqrt{3}$ −1

2 $\sqrt{2}$ 2 2 $\sqrt{2}$ 2

□□□ 0 1 1 $\sqrt{3}$ ∞ $-\sqrt{3}$ −1 −1 0

$\sqrt{3}$ $\sqrt{3}$

□□□□ □□□□ □□□□ □□□□ □□□□ □□□□ □□□□ □□□□

□□□ −1 −1 $-\sqrt{3}$ −1 $-\sqrt{3}$ −1 −1 0

2 $\sqrt{2}$ 2 2 $\sqrt{2}$ 2

□□□ $-\sqrt{3}$ −1 −1 0 1 1 $\sqrt{3}$ 1

2 $\sqrt{2}$ 2 2 $\sqrt{2}$ 2

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$$\cos 11\sqrt{3} \infty -\sqrt{3} -1 -1 0$$

$$\sqrt{3} \sqrt{3}$$

TRIGONOMETRIC IDENTITIES AND EQUATIONS

We should note that most of equations in algebra have a finite number of roots but in the case of trigonometric equations many have an unlimited number.

Example 1:

–1

Solve the equation $\cos \theta = \frac{1}{2}$ for values of θ from 00 to 3600

2

1

The angle whose sine is $\frac{1}{2}$ is 300 , the angles between 00 and 3600 whose sines are

2

1

$\pm$ are illustrated in the diagram below.

2

$\cos \theta = \frac{1}{2}$ is negative only in third and fourth quadrants. Therefore the required roots of the equation in the given range are 1200 and 3300 .

Example 2:

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Solve the equation $2\cos^2 \theta = 1$, for values of θ from 00 to 3600

Solution

$$2\cos^2 \theta = 1$$

$$2\cos^2 \theta - 1 = 0$$

Factorized, we have

$$\cos \theta (2\cos \theta - 1) = 0$$

So, $\cos \theta = 0$ or $2\cos \theta - 1 = 0$

1

$$\cos \theta = 0 \text{ or } \cos \theta = \frac{1}{2}$$

2

θ

If $\cos \theta = 0$, $\theta = 90^\circ$, 270° and $\cos \theta = \frac{1}{2}$, $\theta = 60^\circ$, 300°

θ

Therefore the roots of the equation from 00 to 3600 inclusive are 00 ,

300 , 1500 1800 , 3600

(If both sides of the equation had been divided by $\cos \theta$, we would have lost some of the roots, namely those for which $\cos \theta = 0$)

Trigonometric identities

Example 3:

Shows that $(\cos \theta + \sin \theta)^2 + (\cos \theta - \sin \theta)^2 = 2$

2

$$(\cos \theta + \sin \theta)^2 = (\cos \theta + \sin \theta)(\cos \theta + \sin \theta) = \cos^2 \theta + 2\cos \theta \sin \theta + \sin^2 \theta$$

$$(\cos \theta - \sin \theta)^2 = (\cos \theta - \sin \theta)(\cos \theta + \sin \theta) = \cos^2 \theta - 2\cos \theta \sin \theta + \sin^2 \theta$$

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$$\therefore \cos^2 \theta + 2\cos \theta \sin \theta + \sin^2 \theta + \cos^2 \theta - 2\cos \theta \sin \theta + \sin^2 \theta = 2$$

$$2\cos^2 \theta + 2\sin^2 \theta = 2$$

$$2(\cos^2 \theta + \sin^2 \theta) = 2$$

$$\text{Note that: } \cos^2 \theta + \sin^2 \theta = 1$$

$$2 \times 1 = 2$$

$\therefore$ Left hand side is equal to right hand side

Mathematically: L.H.S = R.H.S

Example 4:

1 1

Prove that $\frac{1}{\sec \theta - 1} + \frac{1}{\sec \theta + 1} = 2 \operatorname{cosec} \theta \cot \theta$

$$\frac{1}{\sec \theta - 1} + \frac{1}{\sec \theta + 1}$$

Take the L.H.S

$$\frac{1}{\sec \theta - 1} + \frac{1}{\sec \theta + 1}$$

i.e $\frac{1}{\sec \theta - 1} + \frac{1}{\sec \theta + 1} =$

$$\frac{\sec \theta + 1}{(\sec \theta - 1)(\sec \theta + 1)} + \frac{\sec \theta - 1}{(\sec \theta + 1)(\sec \theta - 1)}$$

$$\frac{2 \sec \theta}{2 \sec^2 \theta - 1}$$

$$= \frac{2 \sec \theta}{2 \sec^2 \theta - 1} = \frac{2 \sec \theta}{2 \sec^2 \theta - 1}$$

$$\frac{1}{\sec \theta}$$

$$= 2 \times \frac{1}{\sec \theta} \times \frac{1}{\sec \theta}$$

1

$$\frac{1}{\sec \theta}$$

$$= 2 \frac{1}{\sec \theta} \left[\frac{1}{\sec \theta} \right]$$

$$\frac{1}{\sec \theta}$$

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$$\frac{1}{\sec \theta}$$

$$= 2 \frac{1}{\sec \theta} \left[\frac{1}{\sec \theta} \times \frac{1}{\sec \theta} \right]$$

$$= 2 \frac{1}{\sec \theta} \times \frac{1}{\sec \theta}$$

$$= 2 \frac{1}{\sec \theta} \times \frac{1}{\sec \theta}$$

$$= 2 \cot \theta \operatorname{cosec} \theta$$

$$= 2 \operatorname{cosec} \theta \cot \theta$$

$$\therefore \text{L.H.S} = \text{R.H.S}$$

EXERCISE

3

1. If $\sin \theta = \frac{3}{5}$, find without using tables or calculators, the values of

5

(a) $\cos \theta$ (b) $\tan \theta$

-8

2. If $\cos \theta = \frac{17}{15}$, and θ is obtuse, find without using table or calculators, the

17

values of (a) $\sin \theta$ (b) $\cot \theta$

Solve the equations, giving values of θ from 0° to 360° inclusive.

3. $\sin^2 \theta + \cos \theta + 1 = 0$

4. $\sin^2 \theta = 3\cos \theta - 1$

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Prove the following identities

1 1

5. $1 - \sin \theta + 1 + \sin \theta = 2 \sin^2 \theta$

$\sin^2 \theta + \cos^2 \theta$

6. $1 - \sin \theta - 1 + \sin \theta = 2\cos \theta$

1 1

7. $-\operatorname{cosec} \theta + \cot \theta = 2\cos \theta$

$\operatorname{cosec} \theta - \cot \theta$

Answers:

4 3

1. (a) $\pm$ (b) $\pm$

5 4

15 -8

2. (a) $\pm$ (b) $\pm$

17 15

3. 270°

4. $45^\circ, 63.4^\circ, 225^\circ, 243.4^\circ$

Worked Examples

1. Solve the equation $1 - \cos \theta = \sin^2 \theta$

We are that $\sin^2 \theta + \cos^2 \theta = 1$, we can now rewrite the equation as follows:

$1 - \cos \theta = 1 - \cos^2 \theta$

$= (1 + \cos \theta)(1 - \cos \theta)$

Either $\cos \theta = 1$ or $1 + \cos \theta = 0$

i.e. $\cos \theta = 1$ or $\cos \theta = -1$

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The solutions are 00 , 900 and 1800

$$\text{NB: } \sin^2 \theta + \cos^2 \theta = 1$$

$$\text{From the equation } 1 - \sin^2 \theta = \cos^2 \theta$$

Substituting

$$1 - \sin^2 \theta + \sin^2 2\theta = 1$$

$$\therefore \sin^2 2\theta - \sin^2 \theta = 1 - 1$$

$$\sin^2 \theta (-\sin^2 \theta - 1) = 0$$

$$\sin^2 \theta = 0 \text{ or } \sin^2 \theta - 1 = 0$$

$$\sin^2 \theta = 0 \text{ or } \sin^2 \theta = 1$$

$$\text{Solve the equation, } 1 - \sin^2 \theta = \sin^2 2\theta$$

$$\text{Note that } \sin^2 2\theta + \cos^2 2\theta = 1, \therefore \sin^2 2\theta = 1 - \cos^2 2\theta$$

$$1 - \cos^2 2\theta = \sin^2 2\theta$$

$$1 - \cos^2 2\theta = 1 - \cos^2 2\theta$$

$$1 - \cos^2 2\theta + \cos^2 2\theta = 1$$

$$\cos^2 2\theta - \cos^2 2\theta = 1 - 1$$

$$\cos^2 2\theta - \cos^2 2\theta = 0$$

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$$\sin^2 \theta (\sin^2 \theta - 1) = 0$$

$$\sin^2 \theta = 0 \text{ or } \sin^2 \theta - 1 = 0$$

$$\sin^2 \theta = 0 \text{ or } \sin^2 \theta = 1$$

$$\theta = 00, 1800 \therefore \sin^2 \theta = 1 \Rightarrow \theta = 900$$

$$\theta = 00, 900, 1800$$

The solutions are 00 , 900 , 1800

1 1

2. Prove that $\csc \theta + \cot \theta = 2 \csc \theta$

$$\csc \theta - \cot \theta$$

Take L.H.S

$$1 \cdot 1 \csc \theta + \cot \theta + \csc \theta - \cot \theta$$

$$+ =$$

$$\csc \theta - \cot \theta \csc \theta + \cot \theta (\csc \theta - \cot \theta)(\csc \theta + \cot \theta)$$

$$2 \csc \theta$$

$$= \csc^2 \theta - \cot^2 \theta$$

$$\text{But } \cot^2 \theta + 1 = \csc^2 \theta$$

$$2 \csc \theta$$

$$= \csc^2 \theta + 1 - \cot^2 \theta$$

$$2\operatorname{cosec} \theta$$

$$= 1$$

$$= 2\operatorname{cosec} \theta$$

$$\therefore \text{L.H.S} = \text{R.H.S}$$

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1 1

3. Prove that $-\operatorname{cosec} \theta + \cot \theta = 2\cot \theta$

$$\operatorname{cosec} \theta - \cot \theta$$

Take L.H.S

$$1 \operatorname{cosec} \theta + \cot \theta - \operatorname{cosec} \theta + \cot \theta$$

$$-\operatorname{cosec} \theta + \cot \theta = (\operatorname{cosec} \theta - \cot \theta)(\operatorname{cosec} \theta + \cot \theta)$$

$$\operatorname{cosec} \theta - \cot \theta$$

$$2\cot \theta$$

$$\operatorname{cosec}^2 \theta - \cot^2 \theta$$

$$2\operatorname{cosec} \theta$$

$$= \cot^2 \theta + 1 - \cot^2 \theta$$

$$2\cot \theta$$

$$= 1$$

$$= 2\cot \theta$$

$$\therefore \text{L.H.S} = \text{R.H.S}$$

THE FORMULAE FOR $\sin(\theta \pm \phi)$, $\cos(\theta \pm \phi)$

There are four identities which have many applications in trigonometry, calculus, coordinate geometry and mechanics. They are:

$$1. \sin(A + B) = \sin A \cos B + \cos A \sin B$$

$$2. \cos(A + B) = \cos A \cos B - \sin A \sin B$$

$$3. \sin(A - B) = \sin A \cos B - \cos A \sin B$$

$$4. \cos(A - B) = \cos A \cos B + \sin A \sin B$$

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Two more identities can be deduced from the above four identities.

THE FORMULAE FOR $\tan(\theta \pm \phi)$

The two identities to be deduced from the four above are:

$$1. \tan(A + B) \quad 2. \tan(A - B)$$

Example 1:

$$\tan A + \tan B$$

Now, let us show that $\tan(A + B) =$

$$\frac{\tan A + \tan B}{1 - \tan A \tan B}$$

Using the formulae for $\sin(A + B)$ and $\cos(A + B)$

$$\sin(A+B)$$

$$\tan(A+B) =$$

$$\cos(A+B)$$

$$\begin{aligned} & \sin A \cos B + \cos A \sin B \\ &= \cos A \cos B - \sin A \sin B \end{aligned}$$

Dividing the numerator and denominator of the right hand side by $\cos A \cos B$.

$$\begin{aligned} & \frac{\sin A \cos B + \cos A \sin B}{\cos A \cos B} \\ & \tan(A+B) = \frac{\sin A \cos B + \cos A \sin B}{\cos A \cos B} \end{aligned}$$

$$\begin{aligned} & \frac{\sin A \sin B}{\cos A \cos B} \\ &= \sin A \sin B \end{aligned}$$

$$1 - x$$

$$\cos A \cos B$$

$$\begin{aligned} & \tan A + \tan B \\ & \therefore \tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B} \end{aligned}$$

$$45$$

Example 2:

$$\begin{aligned} & \tan A - \tan B \\ & \text{Show that } \tan(A-B) = \frac{\tan A - \tan B}{1 + \tan A \tan B} \end{aligned}$$

Using the formulae for $\sin(A-B)$ and $\cos(A-B)$, we obtain

$$\begin{aligned} & \sin A \cos B - \cos A \sin B \\ & \tan(A-B) = \frac{\sin A \cos B - \cos A \sin B}{\cos A \cos B + \sin A \sin B} \end{aligned}$$

$$\begin{aligned} & \sin A \cos B + \cos A \sin B \\ & \cos A \cos B - \sin A \sin B \end{aligned}$$

Dividing the numerator and denominator of the right hand side by $\cos A \cos B$.

$$\begin{aligned} & \frac{\sin A \cos B + \cos A \sin B}{\cos A \cos B - \sin A \sin B} \end{aligned}$$

$$\cos A \cos B$$

$$\tan(A-B) = \frac{\cos A \cos B}{\cos A \cos B}$$

$$\cos A \cos B$$

$$\cos A \cos B$$

$$\sin A \sin B = \sin A \sin B$$

$$+ 1 + x$$

$$\cos A \cos B \cos A \cos B \cos A \cos B$$

$$\tan A - \tan B$$

$$\therefore \tan (A - B) =$$

$$\frac{1 - \tan A \tan B}{1 + \tan A \tan B}$$

Therefore, for convenience, the six identities are as follows:

$$1. \cos (A + B) = \cos A \cos B - \sin A \sin B$$

$$2. \cos (A - B) = \cos A \cos B + \sin A \sin B$$

$$3. \sin (A + B) = \sin A \cos B + \cos A \sin B$$

$$4. \sin (A - B) = \sin A \cos B - \cos A \sin B$$

$$\tan A + \tan B$$

$$5. \tan (A + B) =$$

$$\frac{1 + \tan A \tan B}{1 - \tan A \tan B}$$

$$\tan A - \tan B$$

$$6. \tan (A - B) =$$

$$\frac{1 - \tan A \tan B}{1 + \tan A \tan B}$$

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Example 3: Find, without using tables or calculators, the value of $\sin (1200^\circ + 450^\circ)$, leaving surds in the answer.

Using the formulae for $\sin (A + B)$

$$\sin (1200^\circ + 450^\circ) = \sin 1200^\circ \cos 450^\circ + \cos 1200^\circ \sin 450^\circ$$

$$\frac{\sqrt{3}}{2}$$

$$\text{Note that } \sin 1200^\circ = \sin 600^\circ =$$

$$\frac{1}{2}$$

$$-\frac{1}{2}$$

$$\cos 1200^\circ = -\cos 600^\circ =$$

$$\frac{1}{2}$$

$$-\frac{1}{2}$$

$$\cos 1200^\circ = -\cos 600^\circ =$$

$$\frac{1}{2}$$

$$\frac{1}{2} \cdot \frac{\sqrt{2}}{2}$$

$$\cos 450^\circ = \cos 45^\circ = \frac{\sqrt{2}}{2}$$

$$\frac{\sqrt{2}}{2}$$

$$\frac{\sqrt{3}}{2} \cdot \frac{\sqrt{2}}{2} - \left(-\frac{1}{2}\right) \cdot \frac{\sqrt{2}}{2}$$

$$\therefore \sin (1200^\circ + 450^\circ) = \frac{\sqrt{3}\sqrt{2}}{4} + \frac{\sqrt{2}}{4}$$

$$\frac{\sqrt{3}\sqrt{2} + \sqrt{2}}{4}$$

$$\frac{\sqrt{2}(\sqrt{3} + 1)}{4}$$

$$= \frac{\sqrt{2}}{4}(\sqrt{3} + 1)$$

$$\frac{\sqrt{2}}{4}(\sqrt{3} + 1)$$

$$\frac{\sqrt{2}}{4}(\sqrt{3} + 1)$$

$$= \frac{\sqrt{2}}{4}(\sqrt{3} + 1)$$

$$\sqrt{2}$$

$$\therefore \sin(1200 + 450) = (\sqrt{3} - 1)$$

$$4$$

Example 4: Find, without using tables or calculators, the value of $\cos(450 - 300)$, leaving surds in the answer.

Using the formulae for $\cos(A - B)$

$$47$$

$$1 \sqrt{2}$$

$$\cos 450 = \sin 450 =$$

$$\sqrt{2} 2$$

$$\sqrt{3} 1$$

$$\cos 300 = , \sin 300 =$$

$$2 2$$

$$\therefore \cos(450 - 300) = \cos 450 \cos 300 + \sin 450 \sin 300$$

$$\sqrt{2} \sqrt{3} \sqrt{2} 1$$

$$= 2 \times 2 + 2 \times 2$$

$$\sqrt{2} \times \sqrt{3} \sqrt{2}$$

$$= +$$

$$4 4$$

$$\sqrt{2}$$

$$= 4(\sqrt{3} + 1)$$

$$\sqrt{2}$$

$$\therefore \cos(450 + 300) = (\sqrt{3} + 1)$$

$$4$$

SELF - ASSESSMENT EXERCISE

Find the values of the following, leaving surds in the answers.

$$1. \sin(300 + 450) \quad 2. \sin(600 + 450) \quad 3. \cos(1200 + 450)$$

$$4. \cos 1050 \quad 5. \sin 1650$$

Example 5: If $\tan(\theta + 450) = 2$, find the value of $\tan \theta$

Note that $\tan 450 = 1$

Using the formulae for $\tan(A + B)$, we obtain

$$48$$

$$\tan \theta + \tan 450$$

$$\tan(\theta + 450) = 2$$

$$1 - \tan \theta \tan 450$$

$$\tan \theta + 1$$

$$= 2$$

$$1 - \tan \theta$$

$$\tan \theta + 1 = 2$$

$$\therefore 1 - \tan \theta = 1$$

$$2(1 - \tan \theta) = \tan \theta + 1$$

$$2 - 2 \tan \theta = \tan \theta + 1$$

$$2 - 1 = 2 \tan \theta + \tan \theta$$

$$\therefore 1 = \tan \theta (2 + 1)$$

$$1 = 3 \tan \theta$$

$$1$$

$$\therefore \tan \theta =$$

$$3$$

EXERCISE

1. If $\tan(\theta + 225^\circ) = 4$, find the value of $\tan \theta$

$$1$$

2. If $\tan(A + B) =$ and $\tan A = 3$, find the value of $\tan B$

$$7$$

THE DOUBLE ANGLE IDENTITIES

The compound angle formulae deal with any two angles A and B and can therefore be used for two equal angles ($B = A$)

Replacing B by A in the compound angle formulae for (A+B) gives.

$$49$$

$$1. \cos 2A \equiv 2 \cos^2 A - 1$$

$$2. \cos 2A \equiv \cos^2 A - \sin^2 A$$

$$2 \tan A$$

$$3. \tan 2A =$$

$$1 - \sin^2 A$$

We can express the second of the above group in several forms e.g:

$$\cos^2 A - \sin^2 A \equiv (1 - \sin^2 A) - \sin^2 A \equiv 1 - 2\sin^2 A$$

$$\cos^2 A - \sin^2 A \equiv \cos^2 A - (1 - \cos^2 A) \equiv 2\cos^2 A - 1$$

This shows that:

$$\cos^2 A - \sin^2 A$$

$$\cos 2A \equiv \{ 1 - 2\sin^2 A$$

$$2\cos^2 A - 1$$

These alternative expressions for $\cos 2A$ can be rearranged to get

$$2\sin^2 A \equiv 1 - \cos 2A$$

$$2\cos^2 A \equiv 1 + \cos 2A$$

THE HALF ANGLE IDENTITIES

$$2 \sin A$$

We know that $\cos^2 2A =$ and, from the previous lesson and also

$$1 - \sin^2 2A$$

$$2 \cos^2 A - 1 = 1 - \sin^2 2A$$

$$\cos^2 2A = \text{and } \sin^2 2A =$$

$$1 + \sin^2 2A = 1 + \cos^2 2A$$

Let us replace $2A$ by A and use t to denote $\sin A$ we have

$$2$$

$$50$$

$$2A$$

$$\cos^2 A =$$

$$1 - \sin^2 A$$

$$2A$$

$$\cos^2 A =$$

$$1 + \sin^2 A$$

$$1 - \sin^2 A$$

$$\cos^2 A =$$

$$1 + \sin^2 A$$

$$1 - \sin^2 2A$$

$$\cos^2 2A =$$

$$1 + \sin^2 2A$$

$$1 - \sin^2 2A$$

Taking R.H.S =

$$1 + \sin^2 2A$$

$$\cos^2 2A$$

$$1 - \sin^2 2A = 1 -$$

$$\cos^2 2A$$

$$= \cos^2 2A$$

$$1 + \sin^2 2A = 1 +$$

$$\cos^2 2A$$

$$\cos^2 2A = \cos^2 2A + \sin^2 2A$$

$$\cos^2 2A = \cos^2 2A$$

$$= \cos^2 A + \sin^2 A$$

$$2 = 1$$

$$\cos^2 2A = \cos^2 2A$$

$$\cos 2A = \cos^2 2A$$

$$= \cos^2 2A \times = \cos 2A$$

$$1$$

$$1 - \sin^2 2A$$

$$\therefore \cos 2A =$$

$$1 + \sin^2 A$$

WORKED EXAMPLES

Prove the following identities.

$$2 \sin^2 A$$

$$1. \cos 2A =$$

$$1 + \sin^2 A$$

$$1 - \sin^2 A$$

$$2. \cos 2A =$$

$$1 + \sin^2 A$$

$$51$$

$$2 \sin^2 A$$

$$1. \cos 2A =$$

$$1 + \sin^2 A$$

$$2 \sin^2 A$$

Taking the R.H.S

$$1 + \sin^2 A$$

$$\sin^2 A$$

$$2$$

$$\sin^2 A$$

$$= \sin^2 A$$

$$1 +$$

$$\sin^2 A$$

$$2 \sin^2 A + 2 \sin^2 A$$

$$\sin^2 A + \sin^2 A$$

$$= \sin^2 A + \sin^2 A$$

$$= 1$$

$$\sin^2 A + \sin^2 A$$

$$2 \sin^2 A + 2 \sin^2 A$$

$$= x = 2 \sin^2 A \cos A$$

$$\sin^2 A + 1$$

$$= \sin^2 A$$

$$2 \sin^2 A$$

$$\therefore \cos 2A =$$

$$1 + \sin^2 A$$

ANALYTICAL GEOMETRY OF A STRAIGHT LINE

Let two points, A and B, be (x_1, y_1) and (x_2, y_2) the distance between A and B

We can use the distances in both x-coordinate and y-coordinate or distance from the y - axis is called the x - coordinate of the point.

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From the above diagram, the length $(x_2 - x_1)$ units and the length BC is $(y_2 - y_1)$

Using Pythagoras' theorem

$$AB^2 = AC^2 + BC^2$$

$$= (x_2 - x_1)^2 + (y_2 - y_1)^2$$

$$\therefore AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

This is the distance formula between two points where the x - coordinate and y - coordinate of the two points are given.

Example 1:

Find the distance between the points A(2, 5) and B(5, 9)

In this case, $(x_1 - y_1) = (2, 5)$ and $(x_2 - y_2) = (5, 9)$

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By applying the formula,

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$AB = \sqrt{(5 - 2)^2 + (9 - 5)^2}$$

$$AB = \sqrt{3^2 + 4^2}$$

$$AB = \sqrt{9 + 16}$$

$$AB = \sqrt{25}$$

$$AB = 5$$

SELF - ASSESSMENTS EXERCISE

1. Find the distance between the following pairs of points

- a. (-1, 5) and (-7, -3) b. (-1, -4) and (-6, -16) c. (-4, 5) and (6, 4)
d. (7, 2) and (13, -10)

2. The distance between the points (16, k) and (1, 1) is 17. Find the value of k.

3. The distance between the points (-1, k) and (-7, -3) is 10.

MID - POINT OF TWO GIVEN POINTS

Let the two given points, A and B, have coordinates of (x_1, y_1) and (x_2, y_2) respectively.

$$\frac{x_2 + x_1}{2} \quad \frac{y_2 + y_1}{2}$$

The mid - point of AB is $(\frac{x_2 + x_1}{2}, \frac{y_2 + y_1}{2})$

2 2

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Example 1:

Find the mid - point of the straight line joining the two given points

$(-4, 2)$ and $(2, -9)$.

In this problem $(x_1, y_1) = (-4, 2)$ and $(x_2, y_2) = (2, -9)$

$$(-4) + 2 \cdot 2 + (-9)$$

∴ the mid - point is (,)

$$2 \cdot 2$$

$$-2 - 7 \cdot 1$$

$$= (,) = (-1, -3 \cdot 2)$$

$$2 \cdot 2$$

DIVISION OF A LINE IN A GIVEN RATIO

The coordinates of the point (x_2 , y_2) in the ratio of $m:n$ are therefore given as:

$$x_2 = \frac{m \cdot x_1 + n \cdot x_2}{m+n}$$

$$y_2 = \frac{m \cdot y_1 + n \cdot y_2}{m+n}$$

$$x_1 + y_1 = x_2 + y_2$$

$$x_2 + y_2 = x_1 + y_1$$

When $m = n$; this reduces the formula to (,)

$$2 \cdot 2$$

Example 2:

A line AB joins points with coordinates (2, -1) and (3, 4). A point P is chosen so that AP:PB = 2:1

Find the coordinates of P.

In this example (x_1 , y_1) = (2, -1) and (x_2 , y_2) = (3, 4)

$$m = 2 \text{ and } n = 1$$

Let the coordinates of P be (x , y) then

$$x_2 = \frac{m \cdot x_1 + n \cdot x_2}{m+n} = \frac{2 \cdot 2 + 1 \cdot 3}{2+1} = \frac{4+3}{3} = \frac{7}{3}$$

$$y_2 = \frac{m \cdot y_1 + n \cdot y_2}{m+n} = \frac{2 \cdot (-1) + 1 \cdot 4}{2+1} = \frac{-2+4}{3} = \frac{2}{3}$$

$$x_1 + y_1 = x_2 + y_2$$

$$55$$

$$x_2 + y_2 = x_1 + y_1$$

$$x_2 + y_2 = x_1 + y_1$$

$$x_1 + y_1 = x_2 + y_2$$

$$2 \cdot 1$$

The required point is (2 , 2)

$$3 \cdot 3$$

THE STRAIGHT LINE EQUATION

The equation of a straight line can be expressed in different forms which include the following:

1. Point - slope form
2. Slope - intercept form
3. Two - point form

4. Double - intercept form

5. General form

6. Normal (perpendicular) form.

1. Point - slope form: The equation of a straight line with slope m and which passes through point $A(x_1, y_1)$ is $y - y_1 = m(x - x_1)$

2. Slope - intercept form: The equation of the straight line having slope m and intercept of c on the y - axis is $y = mx + c$

Example: Find the equation of the line passing through the point $(4, 3)$ with slope

1

4

Solution

Let $P(x, y)$ be any other point on each line

(a) Using the formulae, $y - y_1 = m(x - x_1)$

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1

$$y - 3 = (x - 4)$$

4

$$4y - 12 = x - 4$$

$$4y - x - 8 = 0$$

Or

Using the formulae for the equation of straight line in slope - intercept form.

i.e $y = mx + c$

$$y = mx + c$$

1

$$\therefore c = y - mx \text{ (} y = 3, m = 1, x = 4 \text{)}$$

4

$$1 \cdot 4$$

$$= 3 - x$$

$$4 \cdot 1$$

$$= 3 - 1$$

$$= 2$$

1

$\therefore$ the equation of straight line, $y = x + 2$

4

$$4y = x + 8$$

$$4y - x - 8 = 0$$

3. Two - point form: The equation of the straight line through points $A(x_1, y_1)$ and

$$B(x_2, y_2)$$

$$\text{is } y - y_1 = \left(\frac{y_2 - y_1}{x_2 - x_1} \right) (x - x_1)$$

$$x^2 - x^1$$

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Example: Determine the equation of the line passing through points (-4, 1) and (3, -5)

Solution

$$x^2 - x^1$$

Using the formulae, $y - y_1 = (x - x_1)m$, we have

$$x^2 - x^1$$

$$-5 - 1$$

$$y - 1 = (x - (-4))$$

$$3 - (-4)$$

$$-6$$

$$y - 1 = (x + 4)$$

$$7$$

$$-6$$

$$y - 1 = (x + 4)$$

$$7$$

$$7y - 7 = -6(x + 4)$$

$$7y - 7 = -6x - 24$$

$$7y + 6x + 17 = 0$$

4. Double - intercept form: The equation of the straight line whose intercept on the

$$x$$

x - axis and y - axis are (a, 0) and (0, b) is

$$\frac{x}{a}$$

$$+ \frac{y}{b}$$

$$= 1$$

5. General form: The equation of the straight line in general form is normally put in the form $Ax + By + C = 0$, where A, B, C are arbitrary constants.

$$-x - y$$

In this case, slope $m =$ and y - intercept $b =$

$$-1$$

6. Normal (Perpendicular) Form: The normal form of the equation of the straight line is $x \cos \alpha + y \sin \alpha - p = 0$

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where p = length of the perpendicular line from the origin (0, 0) to the line concerned.

α = angle subtended by the perpendicular line with the positive end of the x - axis.

5:1 Conic Section:

A conic section is the locus of a point that moves in a plane so that its distance from a fixed point (focus) in the plane is in constant ratio (eccentricity) to its distance from a fixed straight line (directrix) in the plane. The conic sections are classified into three groups and are distinguished in shape by the value of the eccentricity i.e

When $e = 1$, the shape is a parabola

When $e < 1$, the shape is an ellipse

When $e > 1$, the shape is a hyperbola

5:2 Parabola:

If a point P is always equidistant from a fixed point and a fixed straight line, the locus of the set of points P is called a parabola. Notice that every parabola is symmetrical about the line through the focus which is perpendicular to the directrix. This line is the axis of the parabola

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The point where a parabola crosses its axis is the vertex and the distance between the vertex and the focus is the focal length.

5:3 Equation of parabola:

Let us choose a point $S(a, 0)$ for the focus and the line $x = -a$ or $x + a = 0$ for the directrix.

Assume a point $P(x, y)$ so that

$$PS = PM$$

At point (x_1, y_1) the gradient of the tangent is m

$$m = \frac{y_1}{x_1}$$

$$2m$$

$\therefore$ The equation of the tangent is $y - y_1 = m(x - x_1)$

$$y_1$$

$$2$$

Since $m = \frac{y_1}{x_1}$, the equation can be written as

$$1$$

$$y_1^2 - 4ax_1 = 2m(x - x_1) = 2\frac{y_1}{x_1}(x - x_1)$$

$$y_1^2 - 4ax_1 = 2y_1 - 2y_1$$

$$60$$

$$y_1^2 = 2y_1 + 2ax_1$$

$$y_1^2 = 2y_1(x + a)$$

Example 4:

Find the equation of the tangent to the parabola $y^2 = 8x$ at the point $(2, 4)$

$$y^2$$

$$y^2 = 8x \rightarrow 2y = 8$$

$\square\square$

$\square\square 8 4$

$\therefore = 2\square = \square$

$\square\square$

$\square\square 4 4$

At the point $(2, 4)$, $=\square = 4=1$

$\square\square$

Thus the equation of the tangent is $\square - 4 = (1)(\square - 2)$

$\square - 4 = \square - 2$

$\square = \square + 2$

b. The equation of a normal at point $\square(\square 1, \square 1)$ to the parabola $\square^2 = 4\square\square$

Using the equation of parabola $\square^2 = 4\square\square$, the gradient of a tangent at point

$2\square$

$(\square 1, \square 1)$ is

$\square 1$

$\square\square$

i.e $2y = 4\square$

$\square\square$

$\square\square 2\square$

$\square\square$

$= \square$

$\square\square 2\square$

At point $(\square 1, \square 1)$ the gradient of the tangent is =

$\square\square \square 1$

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Since a normal is always perpendicular to a tangent, therefore, the gradient of the

$-\square 1$

normal is (i.e reciprocate the gradient of the tangent and assign a non negative

$2\square$

sign).

$-\square 1$

$\therefore$ the equation of the normal is $\square - \square 1 = (\square - \square 1)$

$2\square$

$\square\square$

$=\square=1$ i.e $\square\square = \square\square$ or $\square\square^2 = \square\square^2$

$\square\square$

From the above parabola, $\square$ = focus, $\square\square$ = Latus rectum, $\square$ = origin,

$\square\square l$ = directrix. We have

$$\sqrt{(x - x_1)^2 + (y - y_1)^2} = \sqrt{(x - (-x_1))^2 + (y - y_1)^2}$$

$$x + x_1$$

$$\text{Since } x_1^2 = x_1^2$$

$$\therefore (x - x_1)^2 + y^2 = (x + x_1)^2$$

$$62$$

$$x^2 - 2xx_1 + x_1^2 + y^2 = x^2 + 2xx_1 + x_1^2$$

$$\therefore y^2 = 4xx_1$$

So, for a parabola with focus $(x_1, 0)$ and directrix $x = -x_1$, (x, y) is on the parabola, $y^2 = 4xx_1$

The equation $y^2 = 4xx_1$ is the standard form of the equation of parabola.

Note that when $y^2 = 4xx_1$

- i. the focal length is a ,
- ii. the vertex is the origin
- iii. the axis of the parabola is the x - axis
- iv. x is a quadratic function of y

The parametric equations of a parabola are $x = at^2$, $y = 2at$

Note that the vertex, focus, focal length and directrix of any parabola with horizontal or vertical axis can be identified by comparing the equation with that of the standard parabola $y^2 = 4ax$

$$a=0$$

whose vertex is $\{$ whose focus is $x = 0$, $y = 0$ and whose directrix is $x =$

$$x=0$$

$$-x$$

Example 1:

Find the coordinate of the focus, the length of the latus rectum and the equation of the directrix of the parabola $y^2 = 16x$. Sketch the parabola if $y^2 = 16x$, comparing with $y^2 = 4ax$, we deduce

$$63$$

$$4a = 16, a = 4, x = x$$

$$a=4$$

Thus the focus $= (a = 4, y = 0)$ is $(4, 0)$

The latus rectum $4a = 4(4) = 16$

The equation of the directrix $= (x = -a)$ is $x = -4$

Example 2:

$$3$$

Determine the equation of the parabola whose focus is at $(-3, 0)$ and the directrix is

2

3 3

 $x^2 + y^2 = 0$ then $x =$

2 2

3

 $\therefore$ the equation of the parabola is $x^2 = 4ay = 4(1) y = 4y$

2

i.e $x^2 = 4y$

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Example 3:

Sketch the parabola $x^2 = 4(1 - y)$, marking the vertex, the focus, the directrix, the axis of the parabola and focal length.

If $x^2 = 4(1 - y)$, comparing $x^2 = 4ay$

$4a = 4$, $a = 1 - y$, $a = 1$

$\therefore a = 1$, $1 - y = 0 \therefore y = 1$ ($x = 0$)

Thus the vertex is (0, 1)

The focus is (0, 0)

The directrix ($y = 2$) $1 - y = 2$, $1 - y = 2$

$y = 2$

Equation of Tangent and Normal to a Parabola

(a) the equation of tangent at point (x_1, y_1) to the parabola $x^2 = 4ay$

Differentiate the equation $x^2 = 4ay$ as an implicit function

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 $2x = 4a \frac{dy}{dx}$ i.e $2x = 4a \frac{dy}{dx}$ $\frac{dy}{dx} = \frac{x}{2a}$ $4a \frac{dy}{dx} = x$ $= x$ $4a \frac{dy}{dx} = x$

1. the length of the line joining (3, -4) to (-7, 2) is $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

$\sqrt{(2 - (-4))^2 + (-7 - 3)^2}$

$\sqrt{(2 + 4)^2 + (-10)^2}$

$\sqrt{6^2 + 100}$

$\sqrt{36 + 100}$

$\sqrt{136}$

$\sqrt{4 \times 34}$

$\sqrt{4} \times \sqrt{34}$

$2\sqrt{34}$

2. The gradient of the line perpendicular to the join of (-1, 5) and (2, -3) is

$$\frac{2 - (-1)}{-3 - 5} = \frac{3}{-8}$$

$$\text{Gradient} = -\frac{1}{\frac{3}{-8}} = 2 - (-1) = 2 = 4$$

$$\frac{2 - (-1)}{-3 - 5}$$

$$\frac{2 - (-1)}{-3 - 5}$$

3. The midpoint of the line joining (-1, -3) to (3, -5) is

$$\frac{-1 + 3}{2} = \frac{-3 + (-5)}{2}$$

$$\text{Midpoint} = \left[\frac{-1 + 3}{2}, \frac{-3 + (-5)}{2} \right] = [1, -4]$$

$$\frac{-1 + 3}{2} = \frac{-3 + (-5)}{2}$$

$$\frac{-1 + 3}{2} = \frac{-3 + (-5)}{2}$$

$$= [1, -4]$$

$$\frac{-1 + 3}{2} = \frac{-3 + (-5)}{2}$$

$$= (1, -4)$$

The product of the gradients of perpendicular lines is -1 or if one line has a

$$-1$$

gradient m, the gradient of any line perpendicular to its is

$$\frac{1}{m}$$

$$\frac{1}{m}$$

4. The gradient of the line perpendicular to the join of (-1, 5) and (2, -3) is

$$\frac{2 - (-1)}{-3 - 5} = \frac{3}{-8}$$

$$\text{Gradient} = -\frac{1}{\frac{3}{-8}} = -1 - 2 = -3 = 3$$

$$\frac{2 - (-1)}{-3 - 5}$$

$$\frac{2 - (-1)}{-3 - 5}$$

$$-1 - 3 = 3$$

$$\text{Gradient of line perpendicular to} = -8 = 1 \times () =$$

$$8 \times 8$$

$$3$$

$$67$$